Model scraping for comparison and validation

Summary			
Model name:	Mero_1	Population type*:	ICU
Drug*:	meropenem	Model type:	POPPK
Administration route/mode:	IV/ 3h	No. compartments*:	1
Publication			
Title:	Evaluation of Empirical Dosing Regimens for Meropenem in Intensive Care Unit Patients Using Population Pharmacokinetic Modeling and Target Attainment Analysis		
Author(s):	Guohua An, C Buddy Creech, Nan Wu Roger, L Nation, Kenan Gu, Demet Nalbant, Natalia Jimenez-Truque, William Fissell, Stephanie Rolsma. Pratish C Patel, Amy Watanabe, Nicholas Fishbane, Carl M J Kirkpatrick, Cornelia B Landersdorfer, Patricia Winokur		
Other info:	PMID 36622154		
Publication URL*:	https://pmc.ncbi.nlm.nih.gov/articles/PMC9872596/		
Source Study & Dataset			
Source study:	NA		
Inclusion criteria:	NA Exclusion criteria : patients on intermittent dialysis or extracorporeal membrane oxygenation, receiving probenecid, pregnancy at the time of enrolment		
Patient disease(s):	Respiratory disorders(n=41). infections and infestation (n=30) whom 33 included patients with septic shock		
Drug dosing units*:	mg	Number of patients:	114
Plasma conc. units*:	mg/L	No. samples:	370
Daily dosing scheme	Most common : 1000 mg q8h, 500 mg q6h		
Notes/further info:	n=10 continuous renal replacement therapy (CRRT)		

Patient Characteristics (adjust as appropriate)						
Continuous covariate	**Units**	**Median**	**Mean**	**SD**	**Mode**	**IQR**
Age	Y	63				53-71
Total body weight	kg	85				69-107
Lean body weight	kg	57				46-66
CLCR TBW	mL/min	87				50-128
CLCR LBW	mL/min	56				33-86
SOFA	/24	7				4-9

Categorical covariate	**Feature**	**Count (%)**
Sex male	Yes	53
Sepsis	Yes	47
Septic shock	Yes	29
Acute Kidney injury	Yes	38
Mechanical ventilation	Yes	54
Notes/further info:		

Model scraping for comparison and validation

Final Model Structure			
Compartments:	1		
Elimination (e.g. 1st order):	1st		
Estimation algorithm:	FOCEI	Log-transformed?	Yes
BLLOQ handling method (mg/L)	0.1	Dose compartment:	Central
All covariates tested:	Age sex total body weight lean body weight based on equations developed by Janmahasatian et al.. estimated creatinine clearance using Cockcroft-Gault equation presence or absence of CRRT mortality/severity of disease scores (SOFA) sepsis/septic shock mechanical ventilation acute kidney injury		
Covariate inclusion method:	Forward addition step only covariates that decreased the objective function value by more than 3.84 (1 df at $P < 0.05$) and a backward elimination step where only covariates that produced an increase in the objective function value of >6.63 (1 df at $P < 0.01$). Prior biological knowledge on covariate–parameter relationships was also considered in line with the full covariate model approach.		
Covariates included*:	Creatinine clearance calculated with the Cockcroft-Gault formula using total body weight (CLCR.TBW) CRRT on meropenem total clearance (CLT) total bodyweight (TBW) on the volume of distribution (V)		
'Typical' patient for scaling:	CLCR, TBW 87 mL/min TBW 85kg		
Graphical representation / schematic	NA		
Equations	TV_CLT= TV_CLNR + TV_CLR(1-CRRT)+TV CLCRRT(CRRT) TV_CLT = THETA_1 + THETA_2 *(CLCR,TBW/medianCLCR) * (1-CRRT) + THETA_3 *CRRT with CRRT = 0 or 1 TV_V = THETA_4 * (TBW/medianTBW)		
Notes/further info:			

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	1.59			25.2	
THETA_2 (L/h)	3.69			12.1	
THETA_3 (L/h)	2.39			29.2	
THETA_4 L	35.1			10.3	
Between-subject variability (inter-individual variability)					
OMEGA_CLT(%CV)			47.8	27	
OMEGA_V			58.9	55	

Between-occasion variability (inter-occasion variability)					
NA					
Residual error					
Proportional error	0.316			16	
Additive error (mg/L)	0.209			231.7	
Notes/further info:					

** required as a minimum for model replication*

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	Yes, fig 3	Example plasma conc. profiles:	Yes, fig 1, 2
Other goodness-of-fit plots:	Yes, fig S1	Simulated plasma conc. profiles:	Yes, fig 2
Notes/further info:			

Model scraping for comparison and validation

Summary			
Model name:	Mero_2	Population type*:	ICU
Drug*:	Meropenem	Model type:	POPPK
Administration route/mode*:	IV/NA	No. compartments*:	2
Publication			
Title:	Population Pharmacokinetics/Pharmacodynamics and Clinical Outcomes of Meropenem in Critically Ill Patients		
Author(s):	Apinya Boonpeng, Sutep Jaruratanasirikul, Monchana Jullangkoon , Maseetoh Samaeng Thitima Wattanavijitkul , Rungsun Bhurayanontachai, Sutthiporn Pattharachayakul		
Other info:	PMID 36226944		
Publication URL*:	https://pmc.ncbi.nlm.nih.gov/articles/PMC9664862/		
Source Study & Dataset			
Source study:	NA		
Inclusion criteria:	Adults with confirmed or suspected bacterial infection Intravenous meropenem therapy Hospitalization in a medical or surgical ICU Exclusion criteria: requirement of renal replacement therapy, pregnancy or breastfeeding, hypersensitivity to meropenem		
Patient disease(s):	primary infectious site : respiratory		
Drug dosing units*:	mg	Number of patients:	52
Plasma conc. units*:	mg/L	No. samples:	247
Daily dosing scheme	LD 2000 and 1000 q8h (most common)		

Patient Characteristics (adjust as appropriate)						
Continuous covariate	**Units**	**Median**	**Mean**	**SD**	**Mode**	**IQR**
Age	Y	63				48-74
Weight	kg	62				53-70
BMI	Kg/m²	23				20.6-25.4
CLCR-CG	mL/min	44.6				24.2 - 80.7
Serum Albumin	g/dL	2.4				2- 2.9
Categorical covariate	**Feature**		**Count (%)**			
Sex male	Yes		59.6			
Septic shock	Yes		34.6			
Notes/further info:						

Model scraping for comparison and validation

Final Model Structure			
Compartments:	2		
Elimination (e.g. 1st order):	1st order		
Estimation algorithm:	FOCEI	Log-transformed?	Yes
BLLOQ handling method (mg/L)	0.5	Dose compartment:	Central
All covariates tested:	Age sex actual body weight (BW) ideal body weight (IBW) adjusted body weight (ABW) lean body weight (LBW) body mass index APACHE II score. SOFA score creatinine clearance estimated by the Cockcroft-Gault equation (CLCR-CG) and by the Jelliffe equation (CLCR-JEL) estimated glomerular filtration rate (GFR) GFR-MDRD and GFR-CKDEPI acute kidney injury mechanical ventilation support serum albumin fluid balance shock status septic shock		
Covariate inclusion method:	Decrease in OFV of 3.84 (*P* < 0.05 for 1 degree of freedom for forward addition and an increase of OFV by 6.64 (*P* < 0.01 for 1 df) for a backward deletion step.		
Covariates included*:	$CL_{CR\text{-}CG}$ (CL) , serum ALB (V1), SHOCK (V2)		
'Typical' patient for scaling:	$CL_{CR\text{-}CG}$ 50 mL/min, ALB 2.5 g/dL		
Graphical representation / schematic	NA		
Equations	TV_CL = THETA_1* (1 + THETA_2*(CLCR-CG - 50) TV_V1 = THETA_3 (1 + THETA_4 *(ALB-2.5)) TV_V2= THETA_5+ THETA_6 * SHOCK with SHOCK = 0 or 1 TVQ=THETA_7 THETA_1=THETA_CL THETA_2=THETA_CLR-CG THETA_3=THETA_V1 THETA_4=THETA_ALBV1 THETA_5=THETA_V2 THETA_6=THETA_V2SHOCK		
Notes/further info:	Shock= requiring vasopressors or inotropes to maintain a mean arterial pressure greater than 65 mm Hg		

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	5.16			9.8	
THETA_2 (L/h)	0.016			8.9	
THETA_3 (L)	11.1			10.8	
THETA_4 (L)	-0.255			30.5	

THETA_5 (L)	10.3			31.7	
THETA_6 (L)	11.3			43.1	
THETA_7 (L/h)	9.37			38.4	
Between-subject variability (inter-individual variability)					
OMEGA_CL (CV%)			60.5	11.4	1
OMEGA_ V_1			0 (fixed)	NA	
OMEGA_ V_2			47.7	22.1	36.3
OMEGA_ Q			Not estimated	Not estimated	
Between-occasion variability (inter-occasion variability)					
NA					
Residual error					
Proportional error	0.26			8.3	13
Notes/further info:					

** required as a minimum for model replication*

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	Yes, fig 2	Example plasma conc. profiles:	NA
Other goodness-of-fit plots :	Yes, fig 1	Simulated plasma conc. profiles:	NA
Notes/further info:			

Model scraping for comparison and validation

Summary			
Model name:	Mero_3	Population type*:	ICU
Drug*:	meropenem	Model type:	POPPK
Administration route/mode*:	IV/30 min	No. compartments*:	2
Publication			
Title:	Effect of renal clearance and continuous renal replacement therapy on appropriateness of recommended meropenem dosing regimens in critically ill patients with susceptible life-threatening infections		
Author(s):	Raphaël Burger, Monia Guidi, Valérie Calpini, Frédéric Lamoth, Laurent Decosterd, Corinne Robatel, Thierry Buclin, Chantal Csajka, Oscar Marchetti		
Other info:	PMID 30304491		
Publication URL*:	https://pubmed.ncbi.nlm.nih.gov/30304491/		
Source Study & Dataset			
Source study:	*Robatel* et al : https://pubmed.ncbi.nlm.nih.gov/14615469/		
Inclusion criteria:	NA		
Patient disease(s):	Main infectious site: pulmonary		
Drug dosing units*:	mg	Number of patients:	101*
Plasma conc. units*:	mg/L	No. samples:	380 plasma + 80 FD
Daily dosing scheme	0,5, 1 and 2 g q12h and q8h, 1,5g q12h for subjects undergoing CRRT or with impaired renal function (CLCR < 60 mL/min)		
Notes/further info:	CRRT : haemofiltration and haemodiafiltration (CVVHDF) *Data base : - 86 patients in ICU at the Lausanne university hospital : 30 in a prospective TDM study and 56 included in the institutional TDM program - 15 patients from a previous study of Robatel et al, on CVVHDF with rich plasma and filtrate-dialyse sampling (FD)		

Patient Characteristics (adjust as appropriate)						
Continuous covariate	**Units**	**Median**	**Mean**	**SD**	**Mode**	**IQR**
Age	Y	63				49–70
Weight	Kg	72				58–85
Categorical covariate	**Feature**			**Count (%)**		
Sex male	Yes			56		
Sepsis	Yes			49		
Septic shock	Yes			32		
Calculated CL_{CR} 30 to <60 mL/min	Yes			17		
Calculated CL_{CR} 60 to <90 mL/min	Yes			14		
CRRT	Yes			49		
Notes/further info:	external validation see table 2					

Model scraping for comparison and validation

Final Model Structure			
Compartments:	2		
Elimination (e.g. 1st order):	1st		
Estimation algorithm:	FOCEI	Log-transformed?	Yes
BLLOQ handling method:	NA	Dose compartment:	Central
All covariates tested:	Gender age body weight CLCR		
Covariate inclusion method:	A 3.8 (P = 0.05) and 6.6 (P = 0.01) point change for one additional parameter for forward insertion and backward deletion procedures respectively considered statistically significant.		
Covariates included*:	CLCR (CL) CRRT (CL) QFD (CL) BW (V1)		
'Typical' patient for scaling:	median CLCR = 44 mL/min median BW = 72 kg		
Graphical representation / schematic	NA		
Equations	TV_CL = (THETA_1 + THETA_2*QFD)*CRRT + (THETA_3 * (1+ THETA_4 * ((CLCR-44)/44)))*(1-CRRT) With CRRT = 0 or 1 TV_V1= THETA_5 (BW/72)^THETA_6 TV_Q=THETA_7 TV_V2=THETA_8 THETA_1 = TV_CLres THETA_2 = TV_Sc THETA_3 = TV_CLnoCRRT THETA_4 = THETA_CLCR THETA_5 = TV_Vc THETA_6 = THETA_BW		
Notes/further info:	CL_{CR}: CG equation CLres : meropenem residual clearance in CRRT patients; Sc. sieving coefficient : ratio between plasma and filtrate–dialysate concentrations QFD: FD flow in L/h		

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	3.2			12	

THETA_2	0.75			8	
THETA_3 (L/h)	5.9			7	
THETA_4 (%)	0.71			21	
THETA_5 (L)	16			15	
THETA_6 (power)	1.7			19	
THETA_7(L/h)	14			9	
THETA_8 (L)	15			4	
Between-subject variability (inter-individual variability)					
OMEGA_CL (CV%)			40	12	
OMEGA_Sc (CV%)			26	23	
OMEGA_V1(CV%)			51	15	
OMEGA_Q			Not estimated	Not estimated	
OMEGA_V2			Not estimated	Not estimated	
Between-occasion variability (inter-occasion variability)					
NA					
Residual error					
Proportional error	0.52			15	
Additive error (mg/L)	2.7			16	
Notes/further info:					

** required as a minimum for model replication*

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	Yes, fig 1	Example plasma conc. profiles:	NA
Other goodness-of-fit plots:	NA	Simulated plasma conc. profiles:	Yes, table 6-7
Notes/further info:	No supplementary mat		

Model scraping for comparison and validation

Summary			
Model name:	Mero_4	Population type*:	ICU
Drug*:	meropenem	Model type:	POPPK
Administration route/mode:	IV/30 min	No. compartments*:	2
Publication			
Title:	Development of a dosing algorithm for meropenem in critically ill patients based on a population pharmacokinetic/pharmacodynamic analysis		
Author(s):	Lisa Ehmann, Michael Zoller , Iris K MinichmaY , Christina Scharf , Wilhelm Huisinga, Johannes Zander, Charlotte Kloft		
Other info:	PMID 31229669		
Publication URL*:	https://pubmed.ncbi.nlm.nih.gov/31229669/		
Source Study & Dataset			
Source study:	NCT01793012		
Inclusion criteria:	presence of severe infection age ≥ 18 years and therapy with meropenem Exclusion criteria planned hospitalisation < 4 days or meropenem administration > 48 h prior to study start		
Patient disease(s):	Patients with severe infections		
Drug dosing units*:	mg	Number of patients:	48
Plasma conc. units*:	mg/L	No. samples:	1376
Daily dosing scheme	1000 q8h n=47, 7 CRRT patients non analysed		

Patient Characteristics (adjust as appropriate)

Continuous covariate	Units	Median	Mean	SD	Mode	IQR
Age	Y	56				32-70
Weight	Kg	70				47-121
Serum albumin	g/dL	2.8				2.2-3.6
Creatinine clearance measured	mL/min	80				19-171
Creatinine clearance CG	mL/min	80.8				39.4-170
Fibrinogen	mg/dL	340				187-647
Antithrombin	%	74				49-94
Bilirubin	mg/dL	0.8				0.3-7.4
IL-6	pg/mL	88.3				24-1460
CRP	mg/dL	8.9				2.1-32
CD64 Index	-	1.23				0.53-3.73
Lactate	mmol/L	1.5				0.68-3.64
pH	-	7.41				7.30-7.51
HCO3-	mmol/L	27.4				20.5-37

Categorical covariate	Feature	Count (%)
Sex male	Yes	58.5
Notes/further info:	See suppl appendix. different PK parameters if CRRT	

Model scraping for comparison and validation

Final Model Structure			
Compartments:	2		
Elimination (e.g. 1st order):	1st order		
Estimation algorithm:	FOCEI	Log-transformed?	Yes
BLLOQ handling method:	NA	Dose compartment:	Central
All covariates tested:	27 covariates see appendix table A1		
Covariate inclusion method:	Covariates were preselected using graphical, literature and clinical criteria, then included via stepwise forward selection (OFV driven), followed by backward elimination to retain only significant and precise effects.		
Covariates included*:	CLCRCG (CL) body weight (V1) Albumin (V2)		
'Typical' patient for scaling:	CLCRCG = 80.8 ml/min WT = 70 kg ALB = 2.8 g/dL		
Graphical representation / schematic	NA		
Equations	If CLCRCG ≤ 154 : TV_CL_ = THETA_1*(1+ THETA_2*(CLCR_CG-80.8) If CLCRCG ≥ 154 : TV_CL = TV_CL_INF = THETA_3 TV_V1= THETA_5*(WT/70) $^{THETA\_6}$ TV_V2= THETA_7*(1+ THETA_8*(ALB-2.79)) TV_Q = THETA_9 THETA_2 = THETA_CLCRCG_CL THETA_4 = TV_CLCRCG_INF THETA_6 = THETA_ WT_V1 THETA_8 = THETA_ ALB_V2		
Notes/further info:	CLCRCG : Cockcroft–Gault creatinine clearance CL_INF: CL at CLCRCG_INF (=maximum clearance) CLCRCG_INF: CLCRCG value serving as inflection point N=41 : model validated on non CRRT patients		

Parameter Estimates* **(non CRRT)**					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	9.25	8.42–10.1		4.60	
THETA_2	0.00977	0.00800–0.0115		9.20	
THETA_3 (L/h)	15.9	Not estimated		Not estimated	

THETA_4 (mL/min)	154	133-175		6.90	
THETA_5 (L)	7.89	6.05–9.73		11.9	
THETA_6 (power)	0.945	0.637-1.25		16.6	
THETA_7 (L)	16.1	13.8–18.4		7.40	
THETA_8	–0.202	–0.347–0.0572		36.6	
THETA_9 (L/h)	28.4	19.4–37.4		16.1	
Between-subject variability (inter-individual variability)					
OMEGA_ CL (%CV)		13.2-36.5	27.1	19.3	
OMEGA_ V1 (%CV)		20.6-39.8	31.5	14.3	
OMEGA_ V2 (%CV)		9.02-22.2	16.9	14.4	
OMEGA_Q		Not estimated	Not estimated	Not estimated	
Between-occasion variability (inter-occasion variability)					
KAPPA_ CL (%CV)		9.11-15.2	12.5	12	
Residual error					
Proportional error	0.166	0.145-0.187		6.6	
Additive error (mg/L)	0.246	0.106-0.386		29	
Notes/further info:	%CV = $\sqrt{}e(^{IIV^{\wedge}2}$ – 1) * 100				

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	Yes Fig sup A4	Example plasma conc. profiles:	NA
Other goodness-of-fit plots :	Yes Fig sup A4	Simulated plasma conc. profiles:	Yes Fig sup B1 B2
Notes/further info:			

** required as a minimum for model replication*

Model scraping for comparison and validation

Summary			
Model name:	Mero_5	Population type*:	ICU
Drug*:	meropenem	Model type:	POPPK
Administration route/mode:	IV/2h versus >2h	No. compartments*:	2
Publication			
Title:	Comparison of two empirical prolonged infusion dosing regimens for meropenem in patients with septic shock: A two-center pilot study		
Author(s):	Albrecht Eisert , Christian Lanckohr , Janina Frey , Otto Frey , Sebastian G Wicha , Dagmar Horn Bjoern Ellger , Tobias Schuerholz , Gernot Marx Tim-Philipp Simon		
Other info:	PMID 33515688		
Publication URL*:	https://pubmed.ncbi.nlm.nih.gov/33515688/		
Source Study & Dataset			
Source study:	NA		
Inclusion criteria:	Septic shock according to the SEPSIS-1 definition being aged ≥ 18 years necessity for treatment with meropenem determined by the treating physicians. Exclusion criteria : pregnancy, post-organ transplantation status		
Patient disease(s):	Septic shock		
Drug dosing units*:	mg	Number of patients:	32
Plasma conc. units*:	mg/L	No. samples:	512
Daily dosing scheme	one regimen (UHA): 2000mg q8h meropenem for 2 days followed by 1000 mg q8h meropenem (prolonged infusion > 2h) other regimen (UHM): 1000mg q6h meropenem. (infusion time : 2h)		

Patient Characteristics (center 1/center 2)						
Continuous covariate	**Units**	**Median**	**Mean**	**SD**	**Mode**	**IQR**
Age	Y	64/57				56–74/56-64
Body weight	kg	80/87				71–94/80-90
Body height	cm	173/172				169-180/168-178
Creatinine clearance	mL/min	46/82.9				32.2–67.9/39.9–127
SOFA score		10/11				8-13/7-16
SAPS II score		38/38				32-53/30-56
Leukocytes	/nL	14.2/14.4				8.5-21.6/10.4-18.2
Platelets	/nL	150/323				96-271.5/233-394
Procalcitonin	ng/mL	2.42/1.2				0.7-5.4/0.5-6.3
GFR	mL/min	47.3/90				34.2-76.4/44-90
Bilirubin	mg/dL	0.5/0.8				0.3-1.5/0.5-1.5
Haemoglobin	g/dL	8.2/8.3				7.6-8.6/7.8-9.5
Categorical covariate	**Feature**		**Count (%)**			
Sex male	Yes		61			
Notes/further info:	data from two centers : center 1 = UHA (23 patients) / center 2 = UHM (7 patients)					

	main infectious site abdominal (n=7) **10 patients with renal replacement therapy for at least 1 day**

Final Model Structure			
Compartments:	2		
Elimination (e.g. 1st order):	1st		
Estimation algorithm:	FOCE+I or LAPLACE+I (not specified)	Log-transformed?	Yes
BLLOQ handling method (mg/L):	1	Dose compartment:	Central
All covariates tested:	body weight ideal body weight creatinine clearance estimated by the Cockcroft-Gault equation creatinine clearance estimated by the Cockcroft-Gault equation incorporating ideal body weight using Devine's estimation infusion volume		
Covariate inclusion method:	forward inclusion: alpha : 0.05 , backward elimination alpha 0.01		
Covariates included*:	Creatinine clearance estimated by the Cockcroft-Gault equation using ideal body weight (CLCR_CG_IBW)		
'Typical' patient for scaling:	CLCR_CG_IBW = 48.8 mL/min		
Graphical representation / schematic	NA		
Equations	TV_CL = THETA_1 (1 + THETA_2* (CLCR_CG_IBW – 48.8)) TV_V1=THETA_3 TV_V2=THETA_4 TV_Q=THETA_5 THETA_2 = THETA_CLCR_CG_IBW_CL		
Notes/further info:			

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	7.42	6.22-8.24		7.3	
THETA_2 (min/mL)	0.0154	0.0107-0.019		15.1	
THETA_3 (L)	45.8	28.3-62.7		27.3	
THETA_4 (L)	9.18	6-40.7		45.4	
THETA_5 (L/h)	2.21	1.20-9		18.7	
Between-subject variability (inter-individual variability)					
$OMEGA\_CL^2$	0.104			26	
$OMEGA\_V1^2$	0.622			15	
OMEGA_V2	Not estimated				
OMEGA_Q	Not estimated				
Between-occasion variability (inter-occasion variability)					
$KAPPA\_CL^2$	0.083			9.1	

KAPPA_V1[2]	0.0179			35.2	
Residual error					
Proportional error	0.292			4.9	
Notes/further info:					

** required as a minimum for model replication*

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	Yes, fig 3	Example plasma conc. profiles:	NA
Other goodness-of-fit plots:	Yes, fig 1, fig 2	Simulated plasma conc. profiles:	NA
Notes/further info:			

Model scraping for comparison and validation

Summary			
Model name:	Mero_7	Population type*:	ICU
Drug*:	meropenem	Model type:	POPPK
Administration route/mode*:	IV/30 min	No. compartments*:	2
Publication			
Title:	Meropenem Target Attainment and Population Pharmacokinetics in Critically Ill Septic Patients with Preserved or Increased Renal Function		
Author(s):	Matthias Gijsen, Omar Elkayal, Pieter Annaert, Ruth Van Daele, Philippe Meersseman , Yves Debaveye , Joost Wauters , Erwin Dreesen , Isabel Spriet		
Other info:	PMID 35035223		
Publication URL*:	https://pubmed.ncbi.nlm.nih.gov/35035223/		
Source Study & Dataset			
Source study:	NCT03560557		
Inclusion criteria:	Patients with meropenem with severe sepsis or septic shock Exclusion : pregnancy do-not-resuscitate order extracorporeal membrane oxygenation, renal replacement therapy decreased renal function defined as an estimated glomerular filtration rate according to the Chronic Kidney Disease Epidemiology Collaboration equation (eGFRCKD-EPI) <70 mL/min/1.73 m2 on the day of PK sampling		
Patient disease(s):	Respiratory 71%		
Drug dosing units*:	mg	Number of patients:	58
Plasma conc. units*:	mg/L	No. samples:	345
Daily dosing scheme	Main regimen 1000 q8h, second 2000 q8h		

Patient Characteristics (adjust as appropriate)

Continuous covariate	Units	Median	Mean	SD	Mode	IQR
Age	Y	63				55-68
Weight	kg	70				60-79
Daily dose	mg	3000				3-6
SOFA score on ICU admission	/24	9				7-10
SOFA score on day of sampling	/24	9				
APACHE II score	71	20				
mCrCl24h first day of sampling	mL/min/1.73m²	84				64-119
eGFRCKD-EPI first day of sampling	mL/min/1.73m²	104				91-119

Categorical covariate	Feature	Count (%)
Sex male	Yes	69
Shock	Yes	55
ICU mortality	Yes	19
Infectious site pulmonary	Yes	72
Notes/further info:	Mechanical ventilation : 73%	

** required as a minimum for model replication*

Mis en forme : Couleur de police : Automatique, Surlignage

Mis en forme : Couleur de police : Automatique

Model scraping for comparison and validation

Final Model Structure			
Compartments:	2		
Elimination (e.g. 1st order):	1st		
Estimation algorithm:	FOCEI	Log-transformed?	Yes
BLLOQ handling method (mg/L):	0.09	Dose compartment:	Central
All covariates tested:	Age body weight body mass index ideal body weight adjusted body weight serum creatinine estimated creatinine clearance according to the Cockcroft-Gault equation (eCrClCG) eGFR according to the Modification of Diet in Renal Disease equation eGFRCKD-EPI mCrCl24h probability of ARC on the day of sampling according to the ARC predictor 30 fluid balance Acute Physiology and Chronic Health Evaluation II, and Sequential Organ Failure Assessment scores		
Covariate inclusion method:	two-way stepwise covariate modelling procedure (forward inclusion α=0.010; backward elimination α=0.001). Tested covariates selected based on previously published (population) PK studies of meropenem and expert consensus		
Covariates included*:	eCrClCG		
'Typical' patient for scaling:	eCrClCG = 111.7 mL/min		
	NA		
Equations	TVCL = THETA_1 (eCrClCG/111.7) $^{THETA\_2}$ TVV1=THETA_3 TVV2=THETA_4 TVQ =THETA_5 THETA_1 = THETA_CL THETA_2 = THETA_CLCRCG_CL		
Notes/further info:	CV% = $\sqrt{\exp(\omega^2) - 1x100\%}$		

Model scraping for comparison and validation

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	13.7	11.8–15.7		7.4	
THETA_2 (power)	0.64	0.37–0.90		20.9	
THETA_3 (L)	25.5	21.3–29.7		8.5	
THETA_4 (L)	12.4	8.6–16.2		15.8	
THETA_5 (L/h)	8.1	2.9–13.3		32.6	
Between-subject variability (inter-individual variability)					
OMEGA_CL (%CV)		44-716	57.8	12.2	
OMEGA_V1		38.2-75.8	57	16.8	
OMEGA_V2 Correlation between CL and V1	0.185	38.6-110.1	74.4	24.5	
Between-occasion variability (inter-occasion variability)					
BOV on CL	NA				
Residual error					
Proportional error	0.311	0.2134-0.409		17.2	
Additive error (mg/L)	0.147	0.1-0.19		45.6	
Notes/further info:					

** required as a minimum for model replication*

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	Yes, fig 2	Example plasma conc. profiles:	NA yes, fig 1
Other goodness-of-fit plots:	Yes, sup fig S2, S3	Simulated plasma conc. profiles:	NA
Notes/further info:			

Model scraping for comparison and validation

Summary			
Model name:	Mero_8	Population type*:	ICU
Drug*:	Meropenem	Model type:	POPPK
Administration route/mode:	IV/30min	No. compartments*:	2
Publication			
Title:	Acute-on-chronic liver failure alters meropenem pharmacokinetics in critically ill patients with continuous hemodialysis: an observational study		
Author(s):	Jörn Grensemann David Busse Christina König Kevin Roedl Walter Jäger Dominik Jarczak Stefanie Iwersen-Bergmann Carolin Manthey Stefan Kluge Charlotte Kloft Valentin Fuhrmann		
Other info:	PMID 32323030		
Publication URL*:	https://pmc.ncbi.nlm.nih.gov/articles/PMC7176801/		
Source Study & Dataset			
Source study:	NA		
Inclusion criteria:	Receive meropenem for clinical indication Required CVVHD Excluded : patients <18 years, other extracorporeal circuit than CVVHD		
Patient disease(s):	2 variceal haemorrhage. 2 pneumonia 1 hepatorenal syndrome 2 spontaneous bacterial peritonitis 1 urosepsis		
Drug dosing units*:	mg	Number of patients:	19
Plasma conc. units*:	mg/L	No. samples:	180
Daily dosing scheme	1000 q8h		
Notes/further info:	ICU HD continuous liver (ACLFn=8) vs non liver failure (NLF, n=11)		

Patient Characteristics (ACLF/NLF)					
Continuous covariate	**Units**	**Median**	**SD**	**Mode**	**IQR**
Age	Y	59/61			46–68/55–75
Weight	Kg	78/77			60–81/55–86
Height	Cm	175/175			166–180/170–184
APACHE II	/71	30/32			25-40/20-38
SOFA	/24	16 /14			15-19/11-19
Blood flow rate	$mL*h^{-1}$	100 /100			100-200/100-200
Dialysate rate	$mL*h^{-1}$	2000 /2000			2000–2750/2000-2000
Ultrafiltrate	$mL*h^{-1}$	25 /150			0-138/50-200
Dialysate dose	$ml*kg^{-1}*h^{-1}$	31/26			23-37/23-40
PT	%	50 /87			29-61/74-103
Bilirubin	mg/dL	7.4/2.1			4.3-18.7/0.4-8.3
Antithrombin	%	48/71			22-54/56-90
Categorical covariate	**Feature**		**Count %**		
Sex male	Yes		75/72		
Patients receiving norepinephrine ≥ 0.01 $μg*kg^{-1}*min^{-1}$	Yes		88/73		
Notes/further info:					

Model scraping for comparison and validation

Final Model Structure			
Compartments:	2		
Elimination (e.g. 1st order):	1		
Estimation algorithm:	NA	Log-transformed?	Yes
BLLOQ handling method (mg/L):	1	Dose compartment:	Central
All covariates tested:	NA		
Covariate inclusion method:	Likelihood-ratio test		
Covariates included*:	ACLF (acute on chronic liver failure) status (V2)		
'Typical' patient for scaling:	NA		
Graphical representation / schematic	NA		
Equations	TV_CL=THETA_1 TV_V1=THETA_2 TV_V2= THETA_3 x (1-ACLF) + THETA_4 x ACLF With ACLF = 0 or 1 TV_Q= THETA_5		
Notes/further info:			

Parameter Estimates* (ACLF)					
PK parameter (units)	Value (ACLF)	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	5.06			6.7	
THETA_2 (L) (fixed)	8.31				
THETA_3 (L)	35.5			33.3	
THETA_4 (L)	18.5			33.2	
THETA_5 (L/h)	7.23			9.8	
Between-subject variability (inter-individual variability)					
OMEGA_CL (%CV)			29.8	16.8	
OMEGA_ Q			27.6	28.8	
OMEGA_V1			Not estimated	Not estimated	
OMEGA_V2			Not estimated	Not estimated	
Between-occasion variability (inter-occasion variability)					
NA					
Residual error					
Proportional error	0.22			8.4	
Notes/further info:					

** required as a minimum for model replication*

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	NA	Example plasma conc. profiles:	NA
Other goodness-of-fit plots:	NA	Simulated plasma conc. profiles:	Yes, fig 1/2/3
Notes/further info:			

Model scraping for comparison and validation

Summary			
Model name:	Mero_10	Population type*:	ICU
Drug*:	Meropenem	Model type:	POPPK
Administration route/mode:	IV/NA	No. compartments*:	2
Publication			
Title:	Population Pharmacokinetics of Meropenem in Critically Ill Patients Undergoing Continuous Renal Replacement Therapy		
Author(s):	Arantxazu Isla Alicia Rodriguez-Gascon Inaki F. Troconiz Lorea Bueno Maria Angeles Solinis Javier Maynar. Jose Angel Sanchez-Izquierdo and Jose Luis Pedraz		
Other info:	18307371		
Publication URL*:	https://pubmed.ncbi.nlm.nih.gov/18307371/		
Source Study & Dataset			
Source study:	NA		
Inclusion criteria:	treatment with CRRT for >1 day for ≥20 h/day and isolated or expected causative pathogen susceptible to meropenem		
Patient disease(s):	NA		
Drug dosing units*:	mg	Number of patients:	20
Plasma conc. units*:	mg/L	No. samples:	NA
Daily dosing scheme	500/1000/2000 q6 to 8h		
Notes/further info:	Blood + dialysate-ultrafiltrate sampling. CVVH n = 10 CVVHDF n = 10. 7 polytraumatized, 13 septic Pts		

Patient Characteristics					
Continuous covariate	**Units**	**Mean**	**SD**	**Mode**	**Range**
Age	Y	56.2	20.5		21-83
Bodyweight	Kg	73.1	6.9		59-85
CLCR (mL/min)	mL/min	37.4	42.3		0-118
Total protein concentration in plasma	g/dL	4.7	0.9		3.1-6
Albumin concentration in plasma	g/dL	2	0.5		0.9-2.9
Total bilirubin concentration in plasma	g/dL	0.9	0.4		0.3-1.7
Flow	mL/h	1910	616.4		1000-2800
fu		0.79	0.08		0.63-0.98
SOFA score	/24	13.1	4		8-21
APACHE II score	/71	19.4	6.8		7-31
Categorical covariate	**Feature**			**Count %**	
Sex (male)	Yes			75	
CVVH	Yes			50	
Notes/further info:	CLCR = (Ucr *Vu)/(Pcr * t) (24h), *Ucr and Pcr are the creatinine concentrations in urine and plasma, respectively, Vu is the urine volume collected in 24 hours and t is the urinary collection time (1440 min)*				

Mis en forme : Surlignage

Mis en forme : Anglais (Royaume-Uni)

Final Model Structure			
Compartments:	2		
Elimination (e.g. 1st order):	1		
Estimation algorithm:	FOCE +I	Log-transformed?	Yes
BLLOQ handling method (mg/L):	0.25	Dose compartment:	Central
All covariates tested:	Age Bodyweight (BW) Dialysate plus ultrafiltrate flow (FLOW) Creatinine clearance (CLCR) Fu Type of membrane (MEMB; = 1 for AN69 or = 2 for polysulfone) CRRT (= 1 for CVVHDF or = 2 for CVVH) Patient type (1 septic or 2 for polytraumatized)		
Covariate inclusion method:	Significant decrease in -2LL. forward inclusion. backward elimination		
Covariates included*:	CLCr measured in ml/min (CL), patient type (CL , V1)		
'Typical' patient for scaling:	NA		
Graphical representation / schematic	NA		
Equations	TV_CL = THETA_1 + (THETA_2*CLCr) * (2- TYPE) + (THETA_3* CLCR)* (TYPE-1) TV_V1 = THETA_4*(2- TYPE) + THETA_5*(TYPE-1) TYPE = 1 (septic) or 2 (polytraumatized) TV_V2 = THETA_6 TV_Q = THETA_7 THETA_CL =THETA_1 THETA_CLCr_CLseptic =THETA_2 THETA_CLCr_CL_polytraumatized =THETA_3 THETA_V_septic =THETA_4 THETA_V_polytraumatized =THETA_5 Cp = concentrations of meropenem in plasma Cu = concentrations of meropenem in dialysate-ultrafiltrate		

	CLD : distribution clearance Sc : sieving coefficient
Notes/further info:	

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	6.63			13	
THETA_2 (septic) (min/mL)	0.064			41	
THETA_3(polytraumatized) (min/mL)	0.72			21	
THETA_4(septic) (L)	15.7			10	
THETA_5(polytraumatized) (L)	69.5			18	
THETA_6 (L)	19.8			11	
THETA_7 (L/h)	15			14	
Between-subject variability (inter-individual variability)					
OMEGA_ CL (%CV)			38	2	
OMEGA_V1			45 (septic)	35	
OMEGA_V2			Not estimated		
Between-occasion variability (inter-occasion variability)					
NA					
Residual error					
Proportional error	0.21		11		
Additive error (mg/L)	0.22		20		
Notes/further info:					

** required as a minimum for model replication*

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	Yes, fig 2	Example plasma conc. profiles:	Yes, fig 2
Other goodness-of-fit plots:	Yes, fig1	Simulated plasma conc. profiles:	Yes, fig 2
Notes/further info:			

Model scraping for comparison and validation

Summary			
Model name:	Mero_11	Population type*:	ICU
Drug*:	meropenem	Model type:	POPPK
Administration route/mode:	IV/1h	No. compartments*:	1
Publication			
Title:	Population pharmacokinetics and Monte Carlo dosing simulations of meropenem during the early phase of severe sepsis and septic shock in critically ill patients in intensive care units		
Author(s):	Sutep Jaruratanasirikul . Suriyan Thengyai . Wibul Wongpoowarak . Thitima Wattanavijitkul. Kanyawisa Tangkitwanitjaroen . Waroonrat Sukarnjanaset . Monchana Jullangkoon . Maseetoh Samaeng		
Other info:	PMID 25753628		
Publication URL*:	https://pubmed.ncbi.nlm.nih.gov/25753628/		
Source Study & Dataset			
Source study:	NA		
Inclusion criteria:	Adults with diagnosis of severe sepsis or septic shock, either at admission or during the ICU stay Exclusion : pregnancy, hypersensitivity to carbapenems, history of chronic kidney disease		
Patient disease(s):	severe sepsis and septic shock.		
Drug dosing units*:	mg	Number of patients:	9
Plasma conc. units*:	mg/L	No. samples:	171
Daily dosing scheme	1000 q8h		
Notes/further info:	main infectious sites : pulmonary, blood majority of the patients with APACHE II scores of ≥18 and SOFA scores of ≥8		

Patient Characteristics (adjust as appropriate)						
Continuous covariate	**Units**	**Median**	**Mean**	**SD**	**Mode**	**range**
Age	Y		57	16		33-83
Weight	Kg		63	11		49-81
CLcr (MDRD)	mL/min	59.43				12.37-214.55
Categorical covariate	**Feature**			**Count (%)**		
Sex male	Yes			89		
Notes/further info:						

Final Model Structure			
Compartments:	1		
Elimination (e.g. 1^{st} order):	1^{st}		
Estimation algorithm:	FOCEI	Log-transformed?	Yes
BLLOQ handling method (mg/L):	0.08	Dose compartment:	Central
All covariates tested:	Age body weight body mass index systolic blood pressure diastolic blood pressure fluid intake (per day) fluid output (per day) arterial blood pH		

	blood urea nitrogen level SOFA score APACHE II score serum creatinine level serum albumin level creatinine clearance (CLCr) estimated with the Cockcroft-Gault creatinine clearance estimated by Modification of Diet in Renal Disease (MDRD) formula
Covariate inclusion method:	decrease in the minimum objective function value (MOFV) of 3.84 for addition step+ backward deletion (p <0.01)
Covariates included*:	creatinine clearance estimated by Modification of Diet in Renal Disease (MDRD) formula (CL)
'Typical' patient for scaling:	NA
Graphical representation / schematic	NA
Equations	TV_CL = (THETA_1 + THETA_2 * MDRD CLcr) TV_V=THETA_3 THETA_CL =THETA_1 THETA_MDRDCLcr =THETA_2
Notes/further info:	

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	3.01			47.5	
THETA_2 (min/mL)	0.07			32.9	
THETA_3 (L)	23.7			12.6	
Between-subject variability (inter-individual variability)					
OMEGA_ CL (%)			48		
OMEGA_ V			35		
Between-occasion variability (inter-occasion variability)					
NA					
Residual error					
Proportional error	0.2~~.~~				
Additive error (fixed)	0				
Notes/further info:	Combined model error				

** required as a minimum for model replication*

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	NA	Example plasma conc. profiles:	NA
Other goodness-of-fit plots:	Yes, fig 1	Simulated plasma conc. profiles:	NA
Notes/further info:			

Model scraping for comparison and validation

Summary			
Model name:	Mero_12	Population type*:	ICU
Drug*:	meropenem	Model type:	POPPK
Administration route/mode:	IV/20 min	No. compartments*:	2
Publication			
Title:	Dose Optimization of Meropenem in Patients on Veno-Arterial Extracorporeal Membrane Oxygenation in Critically Ill Cardiac Patients: Pharmacokinetic/Pharmacodynamic Modeling		
Author(s):	Soyoung Kang, Seungwon Yang, Jongsung Hahn, June Young, Jang Kyoung, Lok Min, Jin Wi , Min Jung Chang		
Other info:	PMID 36431106		
Publication URL*:	https://pmc.ncbi.nlm.nih.gov/articles/PMC9693387/		
Source Study & Dataset			
Source study:	NCT02581280		
Inclusion criteria:	V-A ECMO and concomitantly receiving meropenem Exclusion : allergy to carbapenems, pregnancy		
Patient disease(s):	myocardial infarction 10/13		
Drug dosing units*:	mg	Number of patients:	13
Plasma conc. units*:	mg/L	No. samples:	NA
Daily dosing scheme	1000 q8h or q12h if CRRT or MDRD<30 mL/min		

Patient Characteristics (adjust as appropriate)						
Continuous covariate	**Units**	**Median**	**Mean**	**SD**	**Mode**	**IQR**
Age	Y	55				53-58
Height	m	1.68				1.63-1.70
Weight	Kg	67.4				57-74.6
eGFR (MDRD)	mL/min	70.4				57.6-101.3
Scr	mg/dL	1.2				0.7–1.56
APACHE II	/71	30				28.5-35
LOS	days	36				25-57.5
Categorical covariate	**Feature**			**Count (%)**		
Sex male	Yes			85		
CRRT	Yes			54		
Notes/further info:	LOS : length of stay					

Model scraping for comparison and validation

Final Model Structure			
Compartments:	2		
Elimination (e.g. 1st order):	1st		
Estimation algorithm:	FOCEI	Log-transformed?	Yes
BLLOQ handling method: (mg/L)	0.1	Dose compartment:	Central
All covariates tested:	Sex age weight height ECMO-associated variables (during ECMO or weaned off of ECMO and ECMO flow rate [LPM. liters per minute]) CRRT-associated variables (use of CRRT, blood flow rate, CRRT 6 h prior to urine output, dialysate flow rate) complete blood count (absolute white blood cells, red blood cells, haemoglobin, and platelets) serum creatinine [SCr] blood urea nitrogen [BUN] Creatinine clearance (CrCL) estimated via the Cockcroft-Gault equation and eGFR estimated via the MDRD equation). alanine transaminase, aspartate aminotransferase, total bilirubin C-reactive protein and procalcitonin blood pressure tympanic body temperature social variables (smoking status and alcohol consumption)		
Covariate inclusion method:	Biological or clinical plausibility. RSE. shrinkage of PK parameters. a condition number of <1000. and visual improvement in the goodness-of-fit plot.		
Covariates included*:	CRRT use (CL)		
'Typical' patient for scaling:	NA		
Graphical representation / schematic	NA		
Equations	$TV\_CL = THETA\_1 * THETA\_2^{CRRT}$ CRRT use =1 , no CRRT=0 TV_V1=THETA_3 TV_V2=THETA_4 TV_Q=THETA_5 THETA_CL =THETA_1 THETA_CRRT_CL =THETA_2		
Notes/further info:			

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	3.79			26	

THETA_2 ~~(power)~~	0.44			30	
THETA_3 (L)	2.4			38	
THETA_4 (L)	8.56			22	
THETA_5 (L/h)	21.3			17	
Between-subject variability (inter-individual variability)					
OMEGA_ CL (CV%)			47.1	49	
OMEGA_V2			44	154	
OMEGA_Q			Not estimated	Not estimated	
Between-occasion variability (inter-occasion variability)					
NA					
Residual error					
exponential error	0.473			21	
Notes/further info:					

** required as a minimum for model replication*

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	Yes, fig 1	Example plasma conc. profiles:	NA
Other goodness-of-fit plots:	Yes, fig S1	Simulated plasma conc. profiles:	NA
Notes/further info:			

Model scraping for comparison and validation

Summary			
Model name:	Mero_13	Population type*:	ICU
Drug*:	meropenem	Model type:	POP
Administration route*:	IV/ 30 min	No. compartments*:	2
Publication			
Title:	Population Pharmacokinetics of Meropenem in Critically Ill Korean Patients and Effects of Extracorporeal Membrane Oxygenation		
Author(s):	Dong-Hwan Lee, Hyoung-Soo Kim, Sunghoon Park, Hwan-il Kim, Sun-Hee Lee, Yong-Kyun Kim		
Other info:	PMID 34834278		
Publication URL*:	https://pmc.ncbi.nlm.nih.gov/articles/PMC8625191/		
Source Study & Dataset			
Source study:	NA		
Inclusion criteria:	NA Exclusion : history of β-lactam allergy or a positive skin test result for meropenem		
Patient disease(s):	Sepsis from unknown source, nosocomial infections, and prophylactic administration for patients undergoing ECMO.		
Drug dosing units*:	mg	Number of patients:	26 (18 non ECMO, 8 ECMO)
Plasma conc. units*:	mg/L	No. samples:	125 (44 samples to validate final model)
Daily dosing scheme	500 or 1000 of meropenem q8 or q12 h		

Patient Characteristics (ECMO/non ECMO)

Continuous covariate	Units	Median	Mean	SD	Mode	IQR
Age	Y	64.0 /72				56.3-66.5/66-80.3
Height	Cm	162 /165				153-169/156-170
Weight	Kg	63.5/54.4				61.9-66.3/50.5-64.5
Body surface area	m^2	1.67 /1.63				1.61-1.75/1.51-1.72
ICU duration	Days	25.0/6.5				6-43.5/4-17.8
APACHE II	/71	21.0 /16 .0				19.5-22.5/12-18
SOFA	/24	9.50/5				8-12.5/3-7.75
BUN	mg/dL	26.9/22.6				22-32/10.8-46.6
Scr	mg/dL	0.820/0.615				0.518-1.15/0.458-1.43
Cystatin C	mg/dL	1.48/1.34				1.43-1.90/0.985-1.86
Albumin	g/dL	3.00/2.55				2.83-3.20/2.30-2.98
Protein	g/dL	5.15/5.05				4.88-5.75/4.7-5.75
CL_{CR} Cockcroft-Gault	mL/min	76.9/73.4				59.5-105/32.7-92.6
GFR. MDRD	mL/min/1.73 m^2	86.9/111				67.1-132/47.5-160
GFR. modified MDRD	mL/min	88.7/95.1				67-115/45-153
GFR. CKD-EPI	mL/min/1.73 m^2	87.7/91.6				70-105/45.6-103
GFR. modified CKD-EPI	mL/min	82.4/77.7				70-94.9/43.6-97
Categorical covariate		**Feature**		**Count (%) ECMO/non ECMO**		
Sex male		Yes		100/78		
CRRT		Yes		37.5/5.5		
ECMO type VA		Yes		88		
Notes/further info:						

Model scraping for comparison and validation

Final Model Structure			
Compartments:	2		
Elimination (e.g. 1st order):	1st		
Estimation algorithm:	FOCEI	Log-transformed?	Yes
BLLOQ handling method (mg/L):	0.2	Dose compartment:	Central
All covariates tested:	Age sex height weight body surface area (BSA) serum albumin level. serum protein level serum creatinine level serum cystatin C level primary diagnosis comorbidity renal function ECMO type (veno–arterial (VA) or veno–venous (VV)) and ECMO flow rate. eGFR, CKD-EPI		
Covariate inclusion method:	stepwise forward inclusion and backward exclusion processes were conducted. Statistical significance was set at $p < 0.01$ ($\Delta OFV < -6.635$ for 1 degree of freedom) for inclusion and $p < 0.001$ ($\Delta OFV > 10.83$ for 1 degree of freedom) for exclusion + clinical significance		
Covariates included*:	CKD-EPI (CL)		
'Typical' patient for scaling:	CKD-EPI = 91.57 ml/min		
Graphical representation / schematic	NA		
Equations	TV_CL = THETA_1 × (1 + THETA_2 *(CE − 91.57)) TV_V1=THETA_3 TV_V2=THETA_4 TV_Q=THETA_5 THETA_CL =THETA_1 THETA_CE_CL =THETA_2		
Notes/further info:	CE :glomerular filtration rate estimated by CKD-EPI equation		

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	6.37			7.41	
THETA_2 (min/mL)	0.00925			10.3	
THETA_3 (L)	9.07			12.2	
THETA_4 (L)	7.91			13.6	
THETA_5 (L/h)	10.7			21.5	

Between-subject variability (inter-individual variability)					
OMEGA_ CL (CV%)			31.4	15.8	3.7
OMEGA_ V1			43.6	22.5	14.7
OMEGA_ V2			36.6	21.0	41.3
OMEGA_Q	Not estimated			Not estimated	
Between-occasion variability (inter-occasion variability)					
BOV on CL					
Residual error					
Proportional error	0.246			29.3	24.2
Power parameter (mg/L~~l~~)	0.865			10	
Notes/further info:					

** required as a minimum for model replication*

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	Yes, fig 2	Example plasma conc. profiles:	Yes, fig S1
Other goodness-of-fit plots:	Yes, fig 1	Simulated plasma conc. profiles:	Yes, fig 3
Notes/further info:			

Model scraping for comparison and validation

Summary			
Model name:	Mero_14	Population type*:	ICU
Drug*:	meropenem	Model type:	POPPK
Administration route/mode*:	IV/3h	No. compartments*:	2
Publication			
Title:	Pharmacokinetics and Monte Carlo Simulation of Meropenem in Critically Ill Adult Patients Receiving Extracorporeal Membrane Oxygenation		
Author(s):	Jae Ha Lee,Dong-Hwan Lee, Jin Soo Kim Won-Beom Jung, Woon Heo, Yong Kyun Kim . Se Hun Kim, Tae-Hoon No, Kyeong Min Jo, Junghae Ko,Ho Young Lee, Kyung Ran Jun, Hye Sook Choi, Ji Hoon Jang Hang-Jea Jang		
Other info:	PMID 34790131		
Publication URL*:	https://pubmed.ncbi.nlm.nih.gov/34790131/		
Source Study & Dataset			
Source study:	NA		
Inclusion criteria:	ECMO for respiratory and/or cardiac dysfunction and who received meropenem for treatment or **prophylaxis**		
Patient disease(s):	Most common disease : pneumonia		
Drug dosing units*:	mg	Number of patients:	30 (10 CRRT, 20 no CRRT)
Plasma conc. units*:	mg/L	No. samples:	210
Daily dosing scheme	**single** dose 500 or 1000		

Patient Characteristics (no CRRT/CRRT)

Continuous covariate	Units	Median	Mean	SD	Mode	IQR
Age (Y)	Y	63.0/63				54-78.5/55.3-65.8
Height (cm)	cm	165 /162				162-172/155-174
Weight (kg)	Kg	67.9 /66.5				56.8-77.5/60.3-86.3
Body surface area	m^2	1.74 /1.70				1.62-1.83/1.63-1.96
ICU duration	days	24.5/29.5				13.5-31.5/16.5-58.3
MV duration	NA	23.5/25				8.5-31.3/15-33.8
SOFA score	/24	10.0/12				7.75-12.3/10.3-13
APACHE II score	/71	26.0/26				23-28.3/23.5-29.8
CRP	mg/dL	8.58/12.9				4.13-18.3/6.61-20.2
Albumin	g/dL	2.75/2.65				2.5-3.1/2.33-2.80
Scr	mg/dL	1.03/1.83				0.71-1.67/1.44-2.32
CL_{CR}. Cockcroft-Gault	mL/min	69.6/43.4				42.2-115/31.8-51
GFR. MDRD	ml/min/ $1.73m^2$	71.8/31				38.1-106/28.5-37.3
GFR. modified MDRD	mL/min	73.4/32.7				42.1-99.7/30.4-37.5
GFR. CKD-EPI	ml/min/ $1.73m^2$	72.9/32.5				38.4-112/29.5-42.8
GFR. modified CKD-EPI	mL/min	77.8/33.6				39.4-102/31.2-43.8
ECO duration	days	15.5/9.0				7.4-20.3/6-13.8
ECMO flow rate	L/min	3.62/3.83				2.62-4.08/3.47-4.17
ECMO	Rev/min	2.868/2.685				2.49-3.03/2.368-2.963

Model scraping for comparison and validation

Categorical covariate	Feature (non CRRT/CRRT)	Count (%)
Sex male	Yes	65/40
pneumonia	Yes	55/50
VA ECMO	Yes	25/20
Notes/further info:	MV : mechanical ventilation, Rev : revolutions	

Final Model Structure			
Compartments:	2		
Elimination (e.g. 1st order):	1st		
Estimation algorithm:	FOCEI	Log-transformed?	Yes
BLLOQ handling method (mg/L):	1	Dose compartment:	Central
All covariates tested:	Sex age weight height body surface area (BSA) serum protein level. serum albumin level primary diagnosis comorbidity Acute Physiology and Chronic Health Evaluation II (APACHE II) score Sequential Organ Failure Assessment Score (SOFA) score presence of CRRT ECMO flow rate (LPM) and ECMO type (veno-arterial or veno-venous). Serum creatinine level and renal function as estimated by applying the Cockcroft-Gault (CG), Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI), modified CKD-EPI Modification of Diet in Renal Disease (MDRD) and modified MDRD equations		
Covariate inclusion method:	Stepwise forward selection and backward elimination statistical significance criteria : $p < 0.01$ (ΔOFV < -6.63 with 1 degree of freedom) for selection and $p < 0.001$ (ΔOFV >10.8 with 1 degree of freedom) for elimination. A significant covariate was required to have clinical relevance and to meet the statistical criteria		
Covariates included*:	CLCR estimated by the Cockcroft-Gault formula (CL) ECMO flow rate (V1)		
'Typical' patient for scaling:	CLCG 49.7 mL/min LPM 3.7 L/min		
Graphical representation / schematic	NA		
Equations	TV_CL = THETA_1 * (1 + THETA_2 * (CG - 49.7)) TV_V1 = THETA_3 * (1 + THETA_4 * (LPM - 3.7)) TV_V2 = THETA_5 TV_Q = THETA_6 THETA_1= THETA_CL THETA_2 =THETA_CLCG_CL THETA_3 = THETA_V1		

	THETA_4=THETA_LPM_V1
Notes/further info:	LPM : ECMO flow rate L/min CG : CG. glomerular filtration rate estimated by Cockcroft-Gault equation

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	7.35			7.33	
THETA_2 (min/mL)	0.0104			30.3	
THETA_3 (L)	17.3			10.9	
THETA_4 (min/L)	0.337			7.10	
THETA_5 (L)	12.8			9.32	
THETA_6 (L/h)	14.5			10.8	
Between-subject variability (inter-individual variability)					
OMEGA_CL (CV%)			39.6	17.4	
OMEGA_V1			48.5	17.1	
OMEGA_Q			Not estimated	Not estimated	
OMEGA_V2			38.8	18.2	
Between-occasion variability (inter-occasion variability)					
NA					
Residual error					
Proportional error	0.0473			7.1	
Additive error (mg/L)	0.37			30.3	
Notes/further info:					

** required as a minimum for model replication*

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	Yes fig 2	Example plasma conc. profiles:	Yes, fig S1
Other goodness-of-fit plots:	Yes, fig 1	Simulated plasma conc. profiles:	Yes, fig 3
Notes/further info:			

Model scraping for comparison and validation

Summary			
Model name:	Mero_16	Population type*:	ICU
Drug*:	meropenem	Model type:	POP
Administration route/mode:	IV/1h	No. compartments*:	2
Publication			
Title:	Population Pharmacokinetic Analysis of Meropenem in Critically Ill Patients With Acute Kidney Injury Treated With Continuous Hemodiafiltration		
Author(s):	Yoko Niibe, Tatsuya Suzuki, Shingo Yamazaki, Takaaki Suzuki, Nozomi Takahashi, Noriyuki Hattori, Taka-Aki Nakada, Shigeto Oda, Itsuko Ishii		
Other info:	PMID 32049890		
Publication URL*:	https://pubmed.ncbi.nlm.nih.gov/32049890/		
Source Study & Dataset			
Source study:	NA		
Inclusion criteria:	Patients who were admitted to the ICU between July 2016 to March 2018 and who were treated with an intravenous infusion of meropenem and were receiving CHDF Exclusion : patients < 20 years		
Patient disease(s):	NA		
Drug dosing units*:	mg	Number of patients:	21
Plasma conc. units*:	mg/L	No. samples:	350
Daily dosing scheme	1000 q12h		

Patient Characteristics (adjust as appropriate)						
Continuous covariate	**Units**	**Median**	**Mean**	**SD**	**Mode**	**range**
Age	Y	70.0				20–81
Weight	Kg	70.6				44–122
Albumin	g/dL	2.2				1.1-3.2
eGFR*	mL/min/1.73 m^2	26.1				5-74.8
Urine output	mL/24 h	167				10-1934
CRP	mg/dL	19.2				2-32
WBC	/10^3 mL	13.4				0.5-39.5
APACHE II score	/71	33				14-47
SOFA score	/24	15				10-23
IL-6*	pg/mL	1807				21-75260
Total CHDF effluent flow rate	mL/h	1500				800-2500
CHDF intensity	mL/kg/h	22				6.6-37.1
Categorical covariate	**Feature**		**Count (%)**			
Sex Male	Yes		43			
CHDF haemofilter	PMMA membrane		76			
Notes/further info:	*On the day of the study					

Mis en forme : Surlignage

Model scraping for comparison and validation

Final Model Structure			
Compartments:	2		
Elimination (e.g. 1st order):	1st		
Estimation algorithm:	FOCEI	Log-transformed?	Yes
BLLOQ handling method (mg/L):	0.5	Dose compartment:	Central
All covariates tested:	Albumin CRP CHDF intensity eGFR urine output APACHE II and SOFA score type of CHDF hemofilter age body weight CRP IL-6 level WBC count sex		
Covariate inclusion method:	forward inclusion (DOFV decrease of more than 3.84 was significant. P< 0.05 for 1 degree of freedom) and backward elimination (DOFV decrease of more than 6.63 was significant. P < 0.01		
Covariates included*:	estimated GFR (CKD EPI)		
'Typical' patient for scaling:	(median value) eGFR CKD EPI = 26.1 ml/min/1.73m^2		
Graphical representation / schematic	NA		
Equations	TV_CL = THETA_1* (eGFR/26.1) $^{THETA\_2}$ TV_V1 = THETA_3 TV_V2 = THETA_4 TV_Q = THETA_5 THETA_1=THETA_CL THETA_2=THETA_eGFR_CL		
Notes/further info:	*CHDF intensity : sum of the dialysate and ultrafiltration rates divided by body weight.*		

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 L/h	4.22			6.8	
THETA_2 (power)	0.25			29.7	
THETA_3 (L)	14.82			6.7	
THETA_4 (L)	11.75			14.3	
THETA_5 (L/h)	7.84			9.5	
Between-subject variability (inter-individual variability)					

OMEGA_CL (%)			26.75	7.5	
OMEGA_V1			27.33	14.1	
OMEGA_Q			Not estimated	Not estimated	
OMEGA_V2			57.05	24.9	
Between-occasion variability (inter-occasion variability)					
BOV on CL	NA				
Residual error					
Proportional error	0.1839			2.2	
Additive error (mg/L)	1.08			47.3	
Notes/further info:					

** required as a minimum for model replication*

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	Yes, fig 2	Example plasma conc. profiles:	NA
Other goodness-of-fit plots:	Yes, fig 1	Simulated plasma conc. profiles:	Yes, fig 3
Notes/further info:			

Model scraping for comparison and validation

Summary			
Model name:	Mero_17	Population type*:	ICU
Drug*:	meropenem	Model type:	POPPK
Administration route/mode:	IV/1h	No. compartments*:	2
Publication			
Title:	Identification of factors affecting meropenem pharmacokinetics in critically ill patients: Impact of inflammation on clearance		
Author(s):	Yoko Niibe, Tatsuya Suzuki, Shingo Yamazaki Masashi Uchida,Takaaki Suzuki, Nozomi Takahashi, Noriyuki Hattori,Taka-Aki Nakada, Itsuko Ishii		
Other info:	PMID 34973877		
Publication URL*:	https://pubmed.ncbi.nlm.nih.gov/34973877/		
Source Study & Dataset			
Source study:	NA		
Inclusion criteria:	Age ≥18 years both medical and surgical origin of ICU admission between July 2016 to September 2017 and treatment with an intravenous infusion of meropenem initiated after ICU admission. Exclusion : if renal replacement therapy		
Patient disease(s):	8/12 : respiratory		
Drug dosing units*:	mg	Number of patients:	12
Plasma conc. units*:	mg/L	No. samples:	237
Daily dosing scheme	1000 q8h		

Patient Characteristics (adjust as appropriate)

Continuous covariate	Units	Median	Mean	SD	Mode	IQR
Age	Y	63				38-76
Weight	Kg	60.6				46-80.5
Albumin	g/dL	2.3				1-3.4
CCr*	mL/min	95.2				53.2-357.3
Urine output	mL/24h	2238.0				905.6-4435
CRP	mg/dL	10.65				1.3-38.3
WBC	$*10^3/\mu L$	13.8				3.4-19.3
APACHE II score	/71	30				16-37
SOFA score	/24	8				1-15
Body temperature	°C	38.0				36.2-40.5
Heart rate		120				80-155

Categorical covariate	Feature	Count (%)
Sex male	Yes	75
Surgery	Yes	58
Microbiology culture positive	Yes	91.7
Notes/further info:	*CCr : Creatinine clearance using the Cockcroft-Gault equation	

Model scraping for comparison and validation

Final Model Structure			
Compartments:	2		
Elimination (e.g. 1st order):	1st		
Estimation algorithm:	FOCEI	Log-transformed?	Yes
BLLOQ handling method (mg/L):	0.5	Dose compartment:	Central
All covariates tested:	Age body weight albumin CRP CCr urine output APACHE II score and SOFA score Sex and patient population (i.e. postoperative or medical patients)		
Covariate inclusion method:	Forward inclusion : decrease of more than 3.84 ($p < 0.05$) in OFV. The covariates were then removed from the model one by one when the increase in OFV was less than 6.63 ($p < 0.01$) during the exclusion		
Covariates included*:	CRP (CL)		
'Typical' patient for scaling:	CRP =10.65 (mg/dL)		
Graphical representation / schematic	NA		
Equations	TV_CL = THETA_1 * (CRP/10.65) $^{THETA\_2}$ TV_V1 = TE~~H~~THA_3 TV_V2 = THETA_4 TV_Q = THETA_5 THETA_1=THETA_CL THETA_2=THETA_CRP_CL		
Notes/further info:			

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 L/h	9.30			4.86	
THETA_2 (power)	-0.14			41.4	
THETA_3 (L)	12.61			10.88	
THETA_4 (L)	7.8			9.97	
THETA_5 (L/h)	9.7			28.87	
Between-subject variability (inter-individual variability)					
OMEGA_CL (CV%)			15.16	4.93	
OMEGA_Q			28.18	21.98	
OMEGA_V1			26.17	12.16	
OMEGA_V2			18.56	17.20	
Between-occasion variability (inter-occasion variability)					
BOV on CL	NA				

Model scraping for comparison and validation

Residual error					
Proportional error	0.2426			10.94	
Additive error (mg/L)	0.6			36.98	
Notes/further info:					

** required as a minimum for model replication*

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	Yes, fig 2	Example plasma conc. profiles:	NA
Other goodness-of-fit plots:	Yes, fig 1	Simulated plasma conc. profiles:	Yes, fig 3
Notes/further info:			

Model scraping for comparison and validation

Summary			
Model name:	Mero_18	Population type*:	ICU
Drug*:	meropenem	Model type:	POPPK
Administration route/mode	IV/NA	No. compartments*:	1
Publication			
Title:	Population Pharmacokinetics and Pharmacodynamics of Meropenem in Critically Ill Patients: How to Achieve Best Dosage Regimen According to the Clinical Situation		
Author(s):	Amaury O'Jeanson Romaric Larcher, Cosette Le Souder, Nassim Djebli, Sonia Khier		
Other info:	PMID 34403127		
Publication URL*:	https://pubmed.ncbi.nlm.nih.gov/34403127/		
Source Study & Dataset			
Source study:			
Inclusion criteria:	Known or suspected infection of a critically ill patient in the medical ICU and treated with meropenem Exclusion : age <18 years		
Patient disease(s):			
Drug dosing units*:	mg	Number of patients:	58 (Whom 26 on RRT)
Plasma conc. units*:	mg/L	No. samples:	115
Daily dosing scheme	1000 or 2000 q6h, 8h or 12h		

Patient Characteristics (adjust as appropriate)

Continuous covariate	Units	Median	Mean	SD	Mode	Range
Age	Y	64	62	14.7		18-85
Weight	Kg	68	72	17.2		39.5-124
BMI	kg/m²	24	25	6.1		15-42
Albumin	g/L	26	26	4.3		15-40
Lactates	mM	1.4	2	1.9		0.5-22.7
Serum creatinine	µM	131	154	109		15-737
Urine creatinine	mM	3.2	3.9	2.9		0.1-17.2
Serum urea	mM	10	13	9.4		0.5-46
Residual diuresis	mL/24h	845	1316	1422		0-7300
CRCLCG	mL/min	52	72	60		10-376
GFR MDRD	mL/min/1.73 m²	49	78	84		7-567

Categorical covariate	Feature	Count (%)
Sex male	Yes	71
RFS = 1 (i.e. GFR > 120 mL/ min/1.73 m2)	Yes	12
RRT	Yes	45
Notes/further info:	Renal function status=RFS	

Model scraping for comparison and validation

Final Model Structure			
Compartments:	1		
Elimination (e.g. 1st order):	1st		
Estimation algorithm:	FOCEI	Log-transformed?	Yes
BLLOQ handling method (mg/L):	0.003	Dose compartment:	Central
All covariates tested:	collected demographic, biologic, and clinical variables		
Covariate inclusion method:	forward selection step (ΔOFV = 3.84, $p < 0.05$) backward deletion step (ΔOFV = 10.8, $p < 0.001$)		
Covariates included*:	eGFR (using the MDRD formula) (CL) presence of a dialysis session (continuous: DIA_C, or semi-continuous: DIA_SC) (CL) renal function status (RFS) and volume of residual diuresis (RD) (CL)		
'Typical' patient for scaling:	RD median 845 mL/24h GFR median 49 mL/min		
Graphical representation / schematic	NA		
Equations	TV_CL = $CL_{RFS=1}$ x RFS + {$CL_{no\ dialysis}$ x (1-DIA_C) x (1-DIA_SC) + CL_{DIA_C}xDIA_C + CL_{DIA_SC} xDIA_SC } x (1-RFS) $TV_{CL} = THETA_1 + THETA_2 \times \left(\frac{RD}{845} - 1\right) \times RFS \times (1 - DIA_C) \times (1 - DIA_{SC}) + \left[\left(THETA_3 + THETA_4 \times CLCR_{MDRD} + THETA_2 \times \left(\frac{RD}{845} - 1\right)\right) \times (1 - DIA_C) \times (1 - DIA_{SC}) + \left(THETA_5 + THETA_2 \times \left(\frac{RD}{845} - 1\right)\right) \times DIA_C + THETA_6 \times DIA_{SC}\right] \times (1 - RFS)$ TV_V = THETA_7		
Notes/further info:	DIA_SC semi-continous and DIA_C continuous = 1 if continuous or semi-continuous dialysis is “on” and = 0 if dialysis is “off” RFS renal function status = 1 if GFR ≥ 120mL/min = *no dialysis needed*. else = 0 = *dialysis needed (continuous or semi-continuous)*		

Mis en forme : Anglais (Royaume-Uni)

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h) *RFS*	13.9			15.5	
THETA_2 *RD*	0.60			31.7	
THETA_3 (L/h) CL	1.36			23.9	
THETA_4 (L.min/h.mL) *CLCR*	0.058			17.8	

THETA_5 (L/h) *DIA_C*	6.38			12.2	
THETA_6 (L/h) *DIA_SC*	11			20.6	
THETA_7 (L)	43.9			17	
Between-subject variability (inter-individual variability)					
OMEGA_CL (CV%)			30.8	37	
OMEGA_V			87.6	31.2	
Between-occasion variability (inter-occasion variability)					
BOV on CL	NA				
Residual error					
Proportional error (CV%)	0.321			19	
Notes/further info:	%CV = √e(IIV^^2 – 1) · 100				

** required as a minimum for model replication*

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	Yes, fig 3	Example plasma conc. profiles:	NA
Other goodness-of-fit plots:	Yes, fig 2 & S1	Simulated plasma conc. profiles:	NA
Notes/further info:			

Model scraping for comparison and validation

Summary			
Model name:	Mero_19	Population type*:	ICU
Drug*:	meropenem	Model type:	POPPK
Administration route/mode:	IV/1h	No. compartments*:	2
Publication			
Title:	Population pharmacokinetics of standard-dose meropenem in critically ill patients on continuous renal replacement therapy: a prospective observational trial		
Author(s):	Dariusz Onichimowski, Anita Będźkowska, Hubert Ziółkowski, Jerzy Jaroszewski, Michał Borys, Mirosław Czuczwar, Paweł Wiczling		
Other info:	PMID 32301057		
Publication URL*:	https://pubmed.ncbi.nlm.nih.gov/32301057/		
Source Study & Dataset			
Source study:	NA		
Inclusion criteria:	Age 18-80 years, medical and surgical origin of ICU admission. treatment with licensed doses of meropenem and clinical indications for CRRT, clinical indication for CRRT due to AKI Exclusion : HIV infection or terminal cancer, intolerance or allergy to beta-lactam, high probability of bacterial infection with meropenem resistant strains, received meropenem 3 months before screened		
Patient disease(s):	NA		
Drug dosing units*:	mg	Number of patients:	19
Plasma conc. units*:	mg/L	No. samples:	256
Daily dosing scheme	1000 q 8 h		

Patient Characteristics (adjust as appropriate)

Continuous covariate	Units	Median	Mean	SD	Mode	Range
Age	Y	67				36–79
Body weight	Kg	80				60–100
Day of antibiotic therapy	days	2				1–12
Albumin concentration	g/L	24.6				15.6–31.8
Creatinine concentration	mg/dL	1.55				0.6–3.7
$eGFR_{MDRD}$ (estimated with MDRD equation)	mL/min	47				15–122
$eGFR_{CG}$ (estimated with Cockroft–Gault equation)	mL/min	50.29				17.5–134.8
Diuresis	mL/h	0				0–90
APACHE	/71	31				8–44
SOFA	/24	10				4–17
Day of filter usage	days	1				1–3
Blood flow	mL/min	160				110–240
Dialysate/substitute flow	mL/h	2800				2100–3500
UF net	L/h	100				0–350

Categorical covariate	Feature	Count (%)
Sex male	Yes	73.7
Septic	Yes	52.6
CVVH	Yes	47.3
CVVHD	Yes	52.6
Notes/further info:	UF net : net ultrafiltration	

Model scraping for comparison and validation

Final Model Structure			
Compartments:	2		
Elimination (e.g. 1st order):	1st		
Estimation algorithm:	FOCEI	Log-transformed?	Yes
BLLOQ handling method (mg/L):	0.1	Dose compartment:	Central
All covariates tested:	Age body weight day of antibiotic therapy presence or absence of sepsis according to Surviving Sepsis Compaigne gender place on the APACHE II and SOFA scale serum albumin level creatinine concentration eGFRMDRD (estimated with the MDRD equation), eGFRCG (estimated with the Cockroft-Gault equation) diuresis parameters of CRRT: type of anticoagulation, day of filter usage, blood flow, dialysate/substitute flow UF net		
Covariate inclusion method:	Difference in OFV between models of 3.84 for 1 degree of freedom was considered to be statistically significant at $p < 0.05$ for the covariate to be included in the base model. backward elimination was performed The least important covariate was dropped from the model according to the OFV unless that difference in OFV was larger than 7.9		
Covariates included*:	Albuminemia (ALB) (V1)		
'Typical' patient for scaling:	ALB 24.6 g/L		
Graphical representation / schematic	NA		
Equations	$TV\_CL = THETA\_1$ $TV\_V1 = THETA\_2*(ALB/24.6)^{THETA\_3}$ $TV\_V2 = THETA\_4$ $TV\_Q = THETA\_5$ THETA_2=THETA_V1 THETA_3=THETA_ALB		
Notes/further info:			

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	15.1			10.1	
THETA_2 (L)	27.9			17.9	
THETA_3 (power)	− 2.87			21.4	
THETA_4 (L)	33.7			28.1	
THETA_5 (L/h)	21.1			16.4	
Between-subject variability (inter-individual variability)					
OMEGA_CL (CV%)			43.7	12.5	1.1
OMEGA_V1			53.1	23.0	8.1
OMEGA_V2			85.6	25.5	30.1
OMEGA_Q			0 fixed	–	

Between-occasion variability (inter-occasion variability)					
NA					
Residual error					
Proportional error	0.241			10.5	
Additive error (mg/L)	0.881			28.4	
Notes/further info:	$\%CV = \sqrt{e(^{IIV^2} - 1)} \cdot 100$				

** required as a minimum for model replication*

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	Yes, fig 3	Example plasma conc. profiles:	Yes, fig 1
Other goodness-of-fit plots:	Yes, fig S1	Simulated plasma conc. profiles:	Yes, fig S2
Notes/further info:			

Model scraping for comparison and validation

Summary			
Model name:	Mero_20	Population type*:	ICU
Drug*:	meropenem	Model type:	POPPK
Administration route/mode:	IV/30 min	No. compartments*:	2
Publication			
Title:	Optimized meropenem dosage regimens using a pharmacokinetic/pharmacodynamic population approach in patients undergoing continuous venovenous haemodiafiltration with high-adsorbent membrane		
Author(s):	A Padullés Zamora, R Juvany Roig, E Leiva Badosa, J Sabater Riera, X L Pérez Fernández, P Cárdenas Campos, R Rigo Bonin, P Alía Ramos, F Tubau Quintano, E Sospedra Martinez, H Colom Codina		
Other info:	PMID 313335959		
Publication URL*:	https://pubmed.ncbi.nlm.nih.gov/31335959/		
Source Study & Dataset			
Source study:	NA		
Inclusion criteria:	Sepsis with acute renal failure requiring CRRT (only CVVHDF) with high-adsorbent membranes for >48 h, isolated or expected causative pathogen susceptible to meropenem Exclusion : hypersensitivity to meropenem or penicillin, neoplasia with metastasis and life expectancy of <6 months, pregnancy or breastfeeding women		
Patient disease(s):	NA		
Drug dosing units*:	mg	Number of patients:	12
Plasma conc. units*:	mg/L	No. samples:	100
Daily dosing scheme	1000 q8h		

Patient Characteristics (adjust as appropriate)						
Continuous covariate	**Units**	**Median**	**Mean**	**SD**	**Mode**	**Range**
Age	Y	70				34-80
Weight	Kg	79				60-110
Height	Cm	167				160-175
Body mass index	kg/m²	28.5				21.3-35.9
SOFA Score	/24	13				9-17
APACHE II Score	/71	29				6-62
Residual diuresis	mL/24h	37				0-200
Categorical covariate	**Feature**			**Count (%)**		
Positive culture	Yes			100		
Sex male	Yes			67		
Notes/further info:	*See Supplemental tables (table S1)*					

Final Model Structure			
Compartments:	2		
Elimination (e.g. 1st order):	1st		
Estimation algorithm:	FOCEI	Log-transformed?	Yes
BLLOQ handling method (mg/L):	0.27	Dose compartment:	Central
All covariates tested:	Weight gender age blood flow or fluid FR		

Covariate inclusion method:	univariate analysis and then by forward inclusion/backward elimination. assuming significance levels of 5% (ΔMOFV = −3.841 units) and 0.1% (ΔMOFV = 10.8 units)
Covariates included*:	No
'Typical' patient for scaling:	NA
Graphical representation / schematic	NA
Equations	No TV_CL = THE~~H~~TA_1 TV_V1 = THETA_2 TV_V2 = THETA_3 TV_Q = THETA_4
Notes/further info:	CLD : distributional CL between central and peripheral compartments = Q

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	7.78			16.45	
THETA_2 (L)	24.9			13.73	
THETA_3 (L)	283			55.12	
THETA_4 (L/h)	6.49			15.72	
Between-subject variability (inter-individual variability)					
OMEGA²_CL	0.258			66.27	
OMEGA²_V1	0.191			34.50	
OMEGA²_V2	Not estimated			Not estimated	
OMEGA²_Q	Not estimated			Not estimated	
Between-occasion variability (inter-occasion variability)					
NA					
Residual error					
Proportional error (sigma²)	0.157			33.25	
Notes/further info:	Sv is the sieving factor. defined as the fraction eliminated across the membrane during CVVHDF, THETA_SV: 0.702 (RSE 7.35), all shrinkage values were <10% for BSV and residual error				

** required as a minimum for model replication*

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	Yes, fig 2	Example plasma conc. profiles:	Yes, fig S1 & 1
Other goodness-of-fit plots:	Yes, fig S2	Simulated plasma conc. profiles:	NA
Notes/further info:			

Model scraping for comparison and validation

Summary			
Model name:	Mero_21	Population type*:	ICU
Drug*:	meropenem	Model type:	POPPK
Administration route/mode:	IV/3min or continuous	No. compartments*:	2
Publication			
Title:	Meropenem dosing in critically ill patients with sepsis and without renal dysfunction: intermittent bolus versus continuous administration? Monte Carlo dosing simulations and subcutaneous tissue distribution		
Author(s):	Jason A. Roberts, Carl M. J. Kirkpatrick, Michael S Roberts, Thomas A. Robertson, Andrew J. Dalley Jeffrey Lipman		
Other info:	PMID 19398460		
Publication URL*:	https://pubmed.ncbi.nlm.nih.gov/19398460/		
Source Study & Dataset			
Source study:			
Inclusion criteria:	Known or suspected sepsis of a critically ill patient and normal renal function (defined as plasma creatinine concentration <120 mmol/L)		
Patient disease(s):	nosocomial pneumonia soft tissue infection, intra-abdominal sepsis and empirical therapy for sepsis without proven source		
Drug dosing units*:	mg	Number of patients:	10
Plasma conc. units*:	mg/L	No. samples:	222
Daily dosing scheme	1000q 8h		

Patient Characteristics (bolus group/ iv continuous group)

Continuous covariate	Units	Median	Mean	SD	Mode	IQR
Age	Y	55/57				48-61/54-63
Total body weight	Kg	80/75				75-85/75-85
Height	Cm	170/175				170-180/173-183
Day 1 SOFA score	/24	3/5				3-4/2-8
Day 2 SOFA score	/24	3/3				2-3/2-4
Plasma creatinine concentration	mmol/L	73/82				55-101/58-112
Creatinine clearance	mL/min	106/93				98-127/69-161

Categorical covariate	Feature	Count (%)
Sex male	Yes	60/75
Outcome (survivors)	Yes	100/60
Notes/further info:	Clearance : CG, randomization 3min vs continous	

Model scraping for comparison and validation

Final Model Structure			
Compartments:	2		
Elimination (e.g. 1st order):	1st		
Estimation algorithm:	FOCEI	Log-transformed?	Yes
BLLOQ handling method (mg/L):	1	Dose compartment:	Central
All covariates tested:	Age Weight body mass index lean body weight sepsis organ failure assessment score plasma creatinine creatinine clearance measured by 8 h urine collection creatinine clearance estimated via Cockcroft–Gault equation (using total body weight) normalized to 6 L/h		
Covariate inclusion method:	Individual empirical bayesian (post hoc) parameters were plotted against covariate values to assess relationships. If a trend between covariates and PK parameter was observed, then it was considered for inclusion in the population model. Possible covariates were added in a stepwise fashion into the model. Covariates were kept in the model if there was improvement in the fit over the base model. i.e. decrease in objective function and decrease in the BSV		
Covariates included*:	CLCR_CG in mL/min(CL)		
'Typical' patient for scaling:	fCG = CG/ 6 (-L/h)		
Graphical representation / schematic	NA		
Equations	TV_CL = THETA_1* fCG TV_CL = THETA_1 x (CLCR/100) THETA_1=THETA_CL TV_V1 = THETA_2 TV_V2 = THETA_3 TV_Q= THETA_4		
Notes/further info:	Cockroft–Gault equation normalized to 6 L/h (fCG)		

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	13.6	12.2-14.9			
THETA_2 (L)	7.9	5.8-10.8			
THETA_3 (L)	14.8	12.5-17.7			
THETA_4 (L/h)	56.3	38.1-78.1			
Between-subject variability (inter-individual variability)					
OMEGA_ CL (CV%)		0.0-25.3	15.3		
OMEGA_V1		0-62.1	44.7		
OMEGA_V2		0.0-23.2	8.4		
OMEGA_Q		0.0 -63.8	22.3		
Between-occasion variability (inter-occasion variability)					
BOV on CL (CV%)		0-16.3	11.8		
BOV on V1		0-43.7	27.7		
BOV on V2		13.2-40.4	28.8		

BOV on Q		0.0-69.1	11.8		
Residual error					
Proportional error	0.152	0.094-0.219			
Additive error (mg/L)	0.43	0.28-0.66			
Notes/further info:					

** required as a minimum for model replication*

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	Yes, fig S2	Example plasma conc. profiles:	Yes, fig 1
Other goodness-of-fit plots:	Yes, fig S1	Simulated plasma conc. profiles:	Yes, fig 2
Notes/further info:			

Model scraping for comparison and validation

Summary			
Model name:	Mero_22	Population type*:	ICU
Drug*:	meropenem	Model type:	POPPK
Administration route/mode:	IV/30 min-1h	No. compartments*:	2
Publication			
Title:	Meropenem pharmacokinetics in critically ill patients with or without burn treated with or without continuous veno-venous haemofiltration		
Author(s):	Daniel J Selig, Kevin S Akers, Kevin K Chung, Kaitlin A Pruskowski Jeffrey R Livezey, Elaine D Por		
Other info:	PMID 34773921		
Publication URL*:	https://pubmed.ncbi.nlm.nih.gov/34773921/		
Source Study & Dataset			
Source study:	NA		
Inclusion criteria:	NA		
Patient disease(s):	NA		
Drug dosing units*:	mg	Number of patients:	23, whom 10 patients with CVVH
Plasma conc. units*:	mg/L	No. samples:	95
Daily dosing scheme	1000 q8h (n=13)		

Patient Characteristic (no burn no CVVH n=8/ no burn CVVH n=4)						
Continuous covariate	**Units**	**Median**	**Mean**	**SD**	**Mode**	**IQR**
Age	Y		39.75/52.5	17.72/16.46		
Admission weight	kg		89.1/90.6	22.05/10.53		
Weight on sampling day	kg		96.29/89.5	17.66/26.75		
Lean body mass on day of sampling	kg		54.9/64.28	12.37/3.56		
Height	cm		159.28 /180.97	22.91/18.25		
Creatinine clearance	mL/min		251.24/105.69	219.08/69.55		
Creatinine*	mg/dL		1.32/1.57 (	1.6/0.78		
Blood urea nitrogen	mg/dL		27.84/29.6	23.27/ 6.52		
Albumin	g/dL		2.43/2.52	0.56/0.25		
Urine output	mL		3439.5/387.25	1633.84/706.18		
Ultrafiltrate flow rate	mL/kg/h		NA/32.7	NA/11.84		
CLCVVH	L/h		NA/2.39	0.49		
Sieving coefficient			NA/0.99	0.02		
Correction factor			NA/0.87	0.02		
Categorical covariate	**Feature**			**Count (%)**		
Sex male	Yes			50/100		
Notes/further info:	*CVVH filters serum creatinine, Burn NO CVVH vs burn CVVH see table 1					

Final Model Structure	
Compartments:	2

Model scraping for comparison and validation

Elimination (e.g. 1st order):	1st		
Estimation algorithm:	FOCEI	Log-transformed?	Yes
BLLOQ handling method (mg/L):	0.025	Dose compartment:	Central
All covariates tested:	Total body weight (WT) lean body mass (LBM) creatinine clearance (CrCl) age total burn surface area (TBSA) in % total second degree burn surface area (SDB) total third degree burn surface area (TDB) serum albumin (ALBUM) urine output (UOP)		
Covariate inclusion method:	Forward addition process. A decrease of at least 3.84 units (α =0.05.df =1) in the objective function value (OFV)		
Covariates included*:	Weight : LBM (CL and V1) CrCL CG (CL) CVVH (CL) Burn (CL) TBSA (V2)		
'Typical' patient for scaling:	CLCRCG = 125 mL/min LBM = 61.7 kg		
Graphical representation / schematic	NA		
Equations	$TVCL = THETA\_1 * (CrCl/125)^{THETA\_2}$ $TVCL\_BURN\_CVVH = THETA\_3 (LBM/61.7)^{THETA\_4} + THETA\_5$ $TVCL\_NO BURN\_CVVH = THETA\_6 * (LBM/61.7)^{THETA\_4} + THETA\_7$ $TV\_CL = \left(\left(THETA\_1 \times \left(\frac{CLCR}{125}\right)^{THETA\_2}\right) \times (1 - CVVH)\right) + CVVH \times \left[\left(\left(THETA\_3 \times \left(\frac{LBM}{61.7}\right)^{THETA_4} + THETA\_5\right) \times BURN\right) + \left(\left(THETA\_6 \times \left(\frac{LBM}{61.7}\right)^{THETA_4} + THETA\_7\right) \times (1 - BURN)\right)\right]$ With BURN = 0 or 1, CVVH = 0 or 1 TV_V1 = THETA_8* (LBM/61.7) TV_V2 = (THETA_9+THETA_10*TBSA) TV_Q = THETA_11 THETA_1=THETA_CL THETA_2=THETA_CRCL_CL THETA_3=THETA_CL_BURN_CVVH THETA_4=THETA_LBM_CL (fixed) THETA_5 = CVVH & Burn (categorical 1+)−0.16		

	THETA_6=THETA_CL_ NO BURN_CVVH THETA_7 = CVVH & no Burn (categorical 1+) −0.58 THETA_8 = V1 THETA_9 = V2 THETA_10 = TBSA on V2 THETA_11 = Q
Notes/further info:	CrCl calculated by the Cockcroft-Gault (C-G) equation LBM calculated using Janmahasatian's formula

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	15.3	9.81–20.79		18.3	
THETA_2 (power)	0.74	0.44-1.03		20.49	
THETA_3 (L/h)	12.85	Not estimated		Not estimated	
THETA_4 (power, fixed)	0.75	NA		NA	
THETA_5	-0.16	-0.63 – 0.31		150	
THETA_6 (power, fixed)	6.43	Not estimated		Not estimated	
THETA_7	-0.58	-0.77—0.39		16.74	
THETA_8 (L)	33.21	22.75–43.67		16.07	
THETA_9 (L)	13.45	4.82–22.07		32/74	
THETA_10 (L)	0.71	0-1.68		69.52	
THETA_11 (L/h)	14.1	3.76–24.43		38.12	
Between-subject variability (inter-individual variability)					
OMEGA²_CL	0.31	0.13–0.49		29.51	
OMEGA²_V1	0.35	0.11–0.58		34.31	
OMEGA²_Q	Not estimated	Not estimated		Not estimated	
OMEGA²_V2	0.61	0.049–1.17		46.94	
Between-occasion variability (inter-occasion variability)					
NA					
Residual error					
Proportional error	0.22	0.14–0.3		18.25	
Additive error (mg/L)	0.0056	0.0011–0.01		40.65	
Notes/further info:	XXXXXX matrix				

Tableau mis en forme

Mis en forme : Couleur de police : Arrière-plan 1

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	Yes, fig S4	Example plasma conc. profiles:	NA
Other goodness-of-fit plots:	Yes, fig 1	Simulated plasma conc. profiles:	Yes, fig 2/3
Notes/further info:			

Model scraping for comparison and validation

Summary			
Model name:	Mero_23	Population type*:	ICU
Drug*:	meropenem	Model type:	POPPK
Administration route/mode:	IV/30 and 5 min	No. compartments*:	2
Publication			
Title:	The combined effects of extracorporeal membrane oxygenation and renal replacement therapy on meropenem pharmacokinetics: a matched cohort study		
Author(s):	Kiran Shekar, John F Fraser, Fabio Silvio Taccone, Susan Welch, Steven C Wallis, Daniel V Mullany, Jeffrey Lipman, Jason A Roberts.		
Other info:	PMID 25636084		
Publication URL*:	https://pubmed.ncbi.nlm.nih.gov/25636084/		
Source Study & Dataset			
Source study:	See doi: 10.1186/1471-2253-12-29 (study protocol), doi: 10.1093/jac/dkp139, doi: 10.1128/AAC.01582-09		
Inclusion criteria:	meropenem during their ECMO therapy Exclusion criteria: Known allergy to study drug pregnancy serum bilirubin concentration >150 µmol/L ongoing massive blood transfusion requirement (>50% blood volume transfused in the previous 8 hours) therapeutic plasma exchange in the preceding 24 hours		
Patient disease(s):	ECMO group: pneumonia, septic shock n=7, cardiogenic shock n=2, sickle-cell crisis n=1, primary graft dysfunction post lung transplant n = 1		
Drug dosing units*:	mg	Number of patients:	21
Plasma conc. units*:	mg/L	No. samples:	NA
Daily dosing scheme	Mainly 1000 mg q8h		
Notes/further info:	Control versus ECMO n=11 ECMO (whom 6 RRT) vs n= 10 non ECMO (=control group) (whom 5 RRT)		

Patient Characteristics (ECMO no RRT/ ECMO RRT)

Continuous covariate	Units	Median	Mean	SD	Mode	IQR
Age	Y	29/38				16-46/23-56
Total body weight	Kg	69/70				60-80/70-76
Day 1 SOFA score		9/16				7-14/13-17
Plasma creatinine concentration	µmol/L	75/NA				44-82/NA
Creatinine clearance	mL/min	108/NA				65-183/NA
Serum bilirubin	µmol/L	23/58				9-73/34-134
Serum albumin	g/L	31/24				27-35/22-32
Serum proteins	g/L	49/44				46-54/34-56
Meropenem daily dose	g	3/3				
Plasma C_{max}	mg/L	42/59				27-56/50-86
Plasma C_{min}	mg/L	4.9/18				2-10/7-43

Categorical covariate	Feature	Count (%)
Sex male	Yes	20/60
Mechanical ventilation	Yes	50/50
Type of ECMO (VV)	Yes	50/40
Notes/further info:	Data from ECMO group	

Model scraping for comparison and validation

Final Model Structure			
Compartments:	2		
Elimination (e.g. 1st order):	1st		
Estimation algorithm:	FOCEI	Log-transformed?	Yes
BLLOQ handling method (mg/L):	1	Dose compartment:	Central
All covariates tested:	Age sex weight serum creatinine concentration Cockroft-Gault-calculated creatinine clearance (CrCL) presence of ECMO and RRT		
Covariate inclusion method:	Stepwise fashion with forward inclusion and backward deletion. A decrease in the OFV of 3.84 units (*P* <0.05) for one degree of freedom was considered statistically significant		
Covariates included*:	CRRT and CLCR-CG (CL)		
'Typical' patient for scaling:	NA		
Graphical representation / schematic	NA		
Equations	TV_CL = THETA_1 x CRRT + (THETA_1 x (THETA_2 x CLCR_CG)) x (1-CRRT) With CRRT = 0 or 1 TV_V1 = THETA_3 TV_V2 = THETA_4 TV_Q = THETA_5		
Notes/further info:			

Commenté [GGC1]: Verifier unite de CG ? plutot en L/H qu'en ml/min ?

Commenté [GGC2R1]: Reunion 10/07/26 avec CL : pas possible dans l'article d'avoir l'unité de theta 2, donc pas possible de savoir quelle est l'unité de CL_CG attendee

A fiare : essayer simulation de dose avec la valuer en mL/min et en L/h pour voir l'allure de la valeur estimée

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	5.1	3,7 – 7,4			
THETA_2 = CL_{CRCL} (fractional)	1.89	0,99 – 2,65			
THETA_3 (L)	18.7	13 – 21			
THETA_4 (L)	13.2	11,3 – 15,9			
THETA_5 (L/h)	21	12,8 – 37			
Between-subject variability (inter-individual variability)					
OMEGA_ CL (%CV)		37,9 – 66,6	51.6		
OMEGA_V1		32,1 – 69,9	45.8		
OMEGA_V2		0,2 – 42,1	28.7		
Between-occasion variability (inter-occasion variability)					
NA					
Residual error					
Proportional error	0.137				
Additive error (mg/L)	2.3				
Notes/further info:					

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	NA	Example plasma conc. profiles:	NA
Other goodness-of-fit plots:	Yes, fig 1	Simulated plasma conc. profiles:	Yes, fig 2
Notes/further info:			

Model scraping for comparison and validation

Summary			
Model name:	Mero_24	Population type*:	ICU
Drug*:	meropenem	Model type:	POPPK
Administration route/mode:	IV/30min	No. compartments*:	1
Publication			
Title:	Meropenem population pharmacokinetics in critically ill patients with septic shock and continuous renal replacement therapy: influence of residual diuresis on dose requirements		
Author(s):	Marta Ulldemolins, Dolors Soy, Mireia Llaurado-Serra, Sergi Vaquer, Pedro Castro, Alejandro H Rodríguez, Caridad Pontes, Gonzalo Calvo, Antoni Torres, Ignacio Martín-Loeches		
Other info:	PMID 26124172		
Publication URL*:	https://pubmed.ncbi.nlm.nih.gov/26124172/		
Source Study & Dataset			
Source study:	NA		
Inclusion criteria:	Adult, Diagnosis of septic shock, CRRT and an indication for treatment with meropenem Exclusion: severe chronic kidney disease requiring RRT		
Patient disease(s):	Main infectious sites : intra-abdominal, respiratory		
Drug dosing units*:	mg	Number of patients:	24
Plasma conc. units*:	mg/L	No. samples:	124
Daily dosing scheme	1000 q8h over 30 min (*n* = 8), 3h infusion (*n* = 5), 500 as 3h infusion n=3		

Patient Characteristics

Continuous covariate	Units	Median	Mean	SD	Mode	Range
Age	Y	68.5				50-81
Weight	Kg	72.8				49-95
APACHE score	/71	26				5-44
SOFA score	/24	12				4-19
No. of accumulated days of meropenem	Day	4				2-22
Total CRRT intensity	mL/kg/h	34.5				18.7-60.1
Dialysate flow rate	mL/h	1				500-1600
Ultrafiltrate flow rate	ml/h	1.2				750-2000
Blood flow	mL/min	200				130-250
Albumin concn	g/L	21.3				12.4-38
Urea concn	mg/dL	64.3				22-168
Creatinine concn	mg/dL	1.6				0.7-2.6
Vol. of diuresis	mL/24 h	76.5				<10-880

Categorical covariate	Feature	Count (%)
Sex male	Yes	50
Patients with hepatic impairment	Yes	20.8
Patients receiving vasopressors	Yes	100
Patients receiving mechanical ventilation	Yes	95.8
Patients receiving CVVHDF/ patients receiving CVVHF	Yes	88/13
Patients surviving	Yes	14
Notes/further info:	*Septic shock by criteria of the Surviving Sepsis Campaign guidelines*	

Model scraping for comparison and validation

Final Model Structure			
Compartments:	1		
Elimination (e.g. 1^{st} order):	1^{st}		
Estimation algorithm:	FOCEI	Log-transformed?	Yes
BLLOQ handling method:(mg/L)	0.4	Dose compartment:	Central
All covariates tested:	All reasonable demographic and clinical variables		
Covariate inclusion method:	Biological plausibility. graphical displays based on the agreement between the observed and predicted drug concentrations. the uniformity of the distribution of the CWRES. improvement of the precision in parameter estimates and the log likelihood ratio test		
Covariates included*:	residual diuresis (CL), body weight (V)		
'Typical' patient for scaling:	Residual diuresis 100 mL weight 73 kg		
Graphical representation / schematic	NA		
Equations	TV_CL = THETA_1 + THETA_2 * (residual diuresis/100) TV_V = THETA_3* $(weight/73)^{THETA\_4}$ THETA_1=THETA_CL THETA_2=THETA_DIUR THETA_3=THETA_V THETA_4=THETA_weight		
Notes/further info:	DIUR : residual diuresis		

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	3.68			11	
THETA_2	0.22			47	
THETA_3 (L)	33.00			10	
THETA_4 (power)	2.07			24	
Between-subject variability (inter-individual variability)					
OMEGA_ CL (CV%)			37	27	
OMEGA_V1			45	61	
Between-occasion variability (inter-occasion variability)					
NA					
Residual error					
Proportional error	-0.258			10	
Additive error (mg/L)	0.0002			42.76	
Notes/further info:					

Model Evaluation Metrics
Does the model publication provide the following?

Visual predictive check (VPC) plot(s):	NA	Example plasma conc. profiles:	NA
Other goodness-of-fit plots:	Yes, fig 1	Simulated plasma conc. profiles:	NA
Notes/further info:	External validation Fig 2		

Summary			
Model name:	Mero_26	Population type*:	ICU
Drug*:	meropenem	Model type:	POPPK
Administration route*:	IV/30 min and continuous	No. compartments*:	1

Publication	
Title:	Meropenem pharmacokinetics in cerebrospinal fluid: comparing intermittent and continuous infusion strategies in critically ill patients—a prospective cohort study
Author(s):	Sebastian G. Wicha, Christina Kinast, Max Münchow, Sandra Wittova, Sebastian Greppmair, Alexandra K. Kunzelmann, Michael Zoller
Other info:	PMID 39082803
Publication URL*:	https://pubmed.ncbi.nlm.nih.gov/39082803/

Source Study & Dataset			
Source study:	NCT04426383		
Inclusion criteria:	Patients with an external ventricular drain treated with meropenem		
Patient disease(s):	Pneumonia (44%), ventriculitis (38%)		
Drug dosing units*:	mg	Number of patients:	16
Plasma conc. units*:	mg/L	No. samples:	243
Daily dosing scheme	2000 q8h (+/- by 8h infusion)		

Patient Characteristics (adjust as appropriate)

Continuous covariate	Units	Median	Mean	SD	Mode	range
Age	Y	59				22-77
Height	cm	175				161-190
Weight	kg	78				62-153
Serum creatinine	µmol/L	61.9				26.5-185.6
Meropenem daily dose	g	6				3-13
Meropenem plasma concentration	mg/L	12.2				0.1-66
Meropenem CSF concentration	mg/L	0.6				0-9
eGFR (CG formula)	mL/min	120				29-248
Bilirubin	µmol/L	5.1				3.4-59.9
C-reactive protein	mg/L	75				2-155
Leukocyte count	G/L	9.6				5.6-15.9
Interleukin-6	pg/mL	18.7				3.5-84.7

Categorical covariate	Feature	Count (%)
Sex male	Yes	81
Primary CNS infection	Yes	56
Infection		
Brain abscess	Yes	6.3
Meningitis	Yes	6.3
Mycotic aneurysm	Yes	6.3
Pneumonia	Yes	44
Ventriculitis	Yes	38
Notes/further info:		

Model scraping for comparison and validation

Final Model Structure			
Compartments:	1		
Elimination (e.g. 1st order):	1st		
Estimation algorithm:	NA	Log-transformed?	Yes
BLLOQ handling method (mg/L):	0.15	Dose compartment:	Central
All covariates tested:	Body weight creatinine clearance according to Cockcroft and Gault (CLCRCG) CSF: continuous administration of meropenem site of infection CSF flow in 24 h total cell count in CSF IL6 (in CSF and plasma) protein (in CSF) lactate (in CSF) C-reactive protein (in plasma) glucose concentration (in CSF)		
Covariate inclusion method:	forward inclusion (Δ objective function value OFV >3.84, P < 0.05; degrees of freedom df = 1) and backward elimination process (Δ OFV >6.63, P < 0.01; df = 1)		
Covariates included*:	Body Weight (CL and V)		
'Typical' patient for scaling:	Body weight = 70 kg		
Graphical representation / schematic	NA		
Equations	$TV\_CL = THETA\_1 * (body\ weight/70)^{0,75}$ $TV\_V = THETA\_2 * (body\ weight/70)$ THETA_1 = THETA_CL THETA_2 = THETA_Vd THETA_3 = THETA_Kp ~~THETA_4 = THETA_KP,IL6~~ ~~THETA_5 = THETA_ FQ~~$\cancel{_{CSF}}$		
Notes/further info:	CSF concentrations measured		

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Model scraping for comparison and validation

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	11.1	9.74-14.3			
THETA_2 (L)	36.2	35.8-39.5			
Between-subject variability (inter-individual variability)					
OMEGA_ CL (CV%)		34.5-64.7	41.2		
OMEGA_V		47.4-55.9	49.9		
Between-occasion variability (inter-occasion variability)					
NA					
Residual error					
Proportional error	0.226	0.224-0.244			
Additive error (mg/L)	0.705	0.59-0.75			
Notes/further info:	equations and covariates for meropenem in CSF available				

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	Yes, fig S2	Example plasma conc. profiles:	Yes, fig 1
Other goodness-of-fit plots:	Yes, fig 2	Simulated plasma conc. profiles:	Yes, fig 3
Notes/further info:			

Summary

Model scraping for comparison and validation

Model name:	Mero_28	Population type*:	ICU
Drug*:	meropenem	Model type:	POPPK
Administration route*:	IV/continuous	No. compartments*:	1
Publication			
Title:	Impact of Continuous Infusion Meropenem PK/PD Target Attainment on C-Reactive Protein Dynamics in Critically Ill Patients With Documented Gram-Negative Hospital-Acquired or Ventilator-Associated Pneumonia		
Author(s):	Carla Troisi, Pier Giorgio Cojutti, Matteo Rinaldi, Tommaso Tonetti, Antonio Siniscalchi, Coen van Hasselt, Pierluigi Viale, Federico Pea		
Other info:	PMID 39455501		
Publication URL*:	https://pubmed.ncbi.nlm.nih.gov/39455501/		
Source Study & Dataset			
Source study:	NA		
Inclusion criteria:	Critically ill patients having documented Gram-negative HAP/VAP Exclusion : antifungal concomitant therapy, mreopenem treatment <48h, pediatric population, intermittent infusion of meropenem		
Patient disease(s):	HAP/VAP		
Drug dosing units*:	mg	Number of patients:	64
Plasma conc. units*:	mg/L	No. samples:	211
Daily dosing scheme	Loading dose 2000 over 2h then initial maintenance dose 1000 q6h or 500 q6h		

Patient Characteristics (adjust as appropriate)

Continuous covariate	Units	Median	Mean	SD	Mode	IQR
Age	Y	65.5				57-74
Height	m	1.70				1.68-1.75
Weight	Kg	75.0				65-85
Serum creatinine	mg/dL	0.96				0.62-1.31
eGFR	mL/min/1.73 m2	70.0				39-100
Meropenem dose	g /24h	2				1.5-4
Treatment duration	days	9				6.5-14
N. of TDM assessment per patient		3				2-4
Meropenem Css	mg/L	15.9				9.8-27.6
Baseline C-RP *	mg/dL	15.7				9.2-26.1
C-RP *	mg/dL	11.5				6.2-20.2
N. of C-RP assessment per patient*		8				5-11
SOFA score*	/24	10				6-15
APACHE score*	/71	17				12-20

Categorical covariate	Feature	Count (%)
Sex male	Yes	56
Patients co-treated with anti-Gram-positive agent	Yes	48.9
Use of mechanical ventilation*	Yes	95.7
Use of ECMO*	Yes	4.3
Presence of shock*	Yes	53.2
Any surgical intervention*	Yes	48.9
Notes/further info:	C-RP : C-Reactive Protein *Calculated on the number of patients included in PD analysis (n = 47)	

Model scraping for comparison and validation

Final Model Structure			
Compartments:	1		
Elimination (e.g. 1^{st} order):	1^{st}		
Estimation algorithm:	SAEM	Log-transformed?	Yes
BLLOQ handling method (mg/L):	NA	Dose compartment:	Central
All covariates tested:	Age sex weight height serum creatinine eGFR		
Covariate inclusion method:	covariate included in the model if reductions in the objective function value (OFV) > 3.84 points, of both the Akaike information criteria (AIC) and Bayesian information criteria (BIC), of the inter-individual variability and of the relative standard errors (RSE) of the fixed-effect parameters observed		
Covariates included*:	eGFR in ml/min/1.73m^2 (CKD-EPI) (CL)		
'Typical' patient for scaling:	NA		
Graphical representation / schematic	NA		
Equations	TV_CL= THETA_1 *e$^{THETA_2×eGFR}$ THETA_1=THETA_CL THETA_2=THETA_eGFR TV_V = THETA_3		
Notes/further info:			

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Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	2.72			16.7	
THETA_2 (power)	0.011			20	
THETA_3 (L)	20 (fixed)				
Between-subject variability (inter-individual variability)					
OMEGA_CL	0.59			10.4	
OMEGA_V	Not estimated			Not estimated	
Between-occasion variability (inter-occasion variability)					
NA					
Residual error					
Proportional error	0.37			6.53	
Notes/further info:	PD model performed				

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	Yes, fig 4	Example plasma conc. profiles:	NA
Other goodness-of-fit plots:	Yes, fig 3	Simulated plasma conc. profiles:	NA
Notes/further info:			

Model scraping for comparison and validation

Summary			
Model name:	Mero_29	Population type*:	ICU
Drug*:	meropenem	Model type:	POPPK
Administration route*:	IV/30 min	No. compartments*:	2
Publication			
Title:	Population pharmacokinetics of four β-lactams in critically ill septic patients comedicated with amikacin		
Author(s):	Isabelle K. Delattre, Flora T. Musuamba , Philippe Jacqmin, Fabio S. Taccone, Pierre-François Laterre, Roger K. Verbeeck, Frédérique Jacobs , Pierre Wallemacq		
Other info:	PMID 22503878		
Publication URL*:	https://pubmed.ncbi.nlm.nih.gov/22503878/		
Source Study & Dataset			
Source study:	NA		
Inclusion criteria:	patients with severe sepsis or septic shock treated with a loading dose of AMK (25 mg/kg) combined with meropenem		
Patient disease(s):	Severe sepsis or septic shock		
Drug dosing units*:	mg	Number of patients:	19
Plasma conc. units*:	mg/L	No. samples:	NA
Daily dosing scheme	1000 q8h		

Patient Characteristics (adjust as appropriate)						
Continuous covariate	**Units**	**Median**	**Mean**	**SD**	**Mode**	**IQR**
NA						
Categorical covariate	**Feature**			**Count (%)**		
NA						
Notes/further info:	Common PKPOP model for meropenem, piperacillin, ceftazidime, cefepime					

Final Model Structure			
Compartments:	2		
Elimination (e.g. 1^{st} order):	1^{st}		
Estimation algorithm:	FOCEI	Log-transformed?	Yes
BLLOQ handling method (mg/L):	NA	Dose compartment:	Central
All covariates tested:	Renal function		
Covariate inclusion method:	Empirical selection		
Covariates included*:	Creatinine clearance (Cockcroft–Gault) (CL) Body weight : (CL, V1, V2, Q)		
'Typical' patient for scaling:	CLCrstd :100 mL/min Body weight : 70 Kg		
Graphical representation / schematic	NA		
Equations	$TV\_CL = THETA\_1 * (CLCri/CLCrstd)^{THETA\_2} \times (BW/70)^{0.75}$ $TV\_V1 = THETA\_3 \times (BW/70)$ $TV\_V2 = THETA\_4 \times (BW/70)$		

	$TV\_Q = THETA\_5 \times (BW/70)^{0.75}$ THETA_1=THETA_CL THETA_2=THETA_CLRF
Notes/further info:	Model 1 RF= CLCri/CLCrstd RF : renal function

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h/70kg)	9.87			15.6	
THETA_2 (power)	0.51			44.9	
THETA_3 (L/70kg)	24.4			12.6	
THETA_4 (L/70kg)	7.01			21.5	
THETA_5 (L/h/70kg)	4.97			18.2	
Between-subject variability (inter-individual variability)					
OMEGA_CL (CV %)			44.1	40.2	
OMEGA_V1 (CV %)			53.6	47	
OMEGA_Q			Not estimated	Not estimated	
OMEGA_V2			Not estimated	Not estimated	
Between-occasion variability (inter-occasion variability)					
NA					
Residual error					
Proportional error	0.371				
Notes/further info:					

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	NA	Example plasma conc. profiles:	NA
Other goodness-of-fit plots:	NA	Simulated plasma conc. profiles:	NA
Notes/further info:			